HISTORY OF CHANGES DURING THE GAP (DoA. Part B)	
CHANGE	**PAGE / SECTION**
DATE	02/10/2020
EVEREST proceeds to sign a withdraw letter	EVEREST is replaced by LAW IN NOVATION (G&S)
Changes in beneficiaries' numbers	EBAN has changed from 17 to 16. EBN has changed from 16 to 17.
DATE	05/10/2020
ACIB declares an amount of 30.400€ as inkind contribution not used on the premises.	It is added at ACIB's budget.
PWR declares an increase of budget needed referred to other direct costs.	The amount needed is transferred from ACIB's to PWR's budget.
DATE	07/10/2020
Section 3.1. The total number of PM is increased.	CEA declares an increase of PM to reflect better the distribution between the people who will be involved in the project.
Changes in subcontracting budget of AITEX.	Adjustment in the distribution of AITEX budget. An amount of Other direct costs budget is transferred to Subcontracting.
Changes in table 3.4	Adjustment in the distribution of Other direct costs.
DATE	13/10/2020
Resources to be committed: the amounts in "Total budget" and "asked funding" should reflect the current situation	Section updated according to the budget.
Changes in table 3.4	Adjustment in the distribution of Other direct costs of AITEX and PARTICULA. Adjustment in the Justification of costs text of each partner.
Table 4.2	Table added.
DATE	19/10/2020
Changes in Table 9. Worpackage tittle table including WP Leader, stat and End month and PM associated	WP2 is AXIA, partner number 19 instead of 18. WP4 starts in month 2 instead of month 1. WP6 starts in month 3 (instead of 4) and ends in month 36 (instead of 28). WP7 starts in month 18 instead of month 24. WP8 leader in PARTI, partner number 15 instead of 14. WP8 starts in month 3 instead of month 1. WP9 leader in EBN partner number 17 instead of 15. WP9 starts in month 12 (instead of month 15), and ends in month 48 (instead of month 40). WP10 starts in month 24 (instead of month 36), and ends in month 48 (instead of month 46).
Section 4.1	Participant description has been included
Section 5.1	This section has been completed.

Associated with document Ref.

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OPEN INNOVATION TEST BED FOR DEVELOPING SAFE NANO-ENABLED BIO-BASED MATERIALS AND POLYMER BIONANOCOMPOSITES
FOR MULTIFUNCTIONAL AND NEW ADVANCED APPLICATIONS

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ACRONYMS		
CAGR: Compound Annual Growth Rate CNC: Cellulose nanocrystals CNF: Cellulose Nanofibres DSC: Differential Scanning Calorimetry. DLS: Dynamic Light Scattering. EDX: Energy Dispersive X-ray. DMA: Dynamic Mechanical Analysis. FTIR: Fourier Transform Infrared. OITB: Open Innovation Test Bed.	GPC: Gel Permeation Chromatography ICP-MS: Inductively Coupled Plasma-Mass Spectrometry NEBM: nano-enabled bio-based materials NMR: Nuclear Magnetic Resonance. NIR: Near Infrared Reflectance. NL: Nanolignina NP: Nanoparticle MFI: Melt Flow Index. OTR: Oxygen Transmission Rate.	PP: Pilot Plant SEM: Scanning Electron Microscope. TEM: Transmission Electron Microscope. TGA: Thermogravimetric Analysis. UV-Vis: Ultraviolet Visible Spectroscopy. WVTR: Water Vapour Transmission Rate. XRD: X-Ray Diffraction. XRF: X-Ray Fluorescence. ºSR: Schopper Riegler Degree

1. Excellence

To succeed in the ambitious objective of achieving a climate-neutral EU by 2050 the bio-based sector, as a key actor, shall respond to relevant challenges in the short term. Climate protection, reduction of greenhouse gas emissions and saving of fossil resources are key elements for a more sustainable future. Biomaterials must offer functional properties for **high-volume applications** and **performing better than fossil-based materials** to be accepted by the industry and end users. The transformation of biobased materials to the nanoscale solves this challenge, providing them with new properties required in many sectors such as a bigger specific surface area, high aspect ratio, and enhanced mechanical, thermal, rheological, optical, barrier and active properties.

BIONANOPOLYS will strengthen the **circularity of nano-enabled bio-based materials (NEBM)** in the economy launching and promoting an **Open Innovation Test Bed (OITB**) to develop **innovative bionanocomposites** from main feedstocks in Europe and **bio-based nanoproducts** in relevant sectors such as: polymers, packaging, textile, agriculture, cosmetics, pharma, hygiene, food and additive manufacturing.

BIONANOPOLYS will improve technologies and processes required in fourteen different pilot lines distributed across Europe within fifteen experienced technological institutes located in ten countries. Four additional partners (technological companies and services providers) will contribute to ensure the feasibility of the test bed, including outreach, dissemination, and a business accelerator programme. Finally, eight key industrial players (SME and LE) will act as final validators in almost twenty different cases studies. In total, BIONANOPOLYS has participation from twelve countries. To ensure the sustainability in long term of the test bed, we need to address the following risks:

› **Acceptance of new technology by the industry:** NEBMs require new technologies and processes. However, still there is a significant uncertainty about its feasibility and market acceptance. Thus, industry do not feel confident to make big investments and there is an additional barrier to attract finance.

› **Seasonal sustainability of feedstocks:** Stable supplies of feedstock in enough quality and quantity are crucial to a successful development from idea to market. Otherwise, the market cannot depend on seasonal fluctuations. › **Price competition and market:** The higher price of bio-based products compared to the fossil-based alternative is a risk. The bio-based products may be confronted with a market that is conservative, sceptic and well-established.

› **Other risks:** The requirements from existing legislation are costly to comply with, particularly for small companies. e.g. nano-ecotoxicology related to the use of nano-enabled materials in industry and/or food contact acceptance.

Considering these challenges, BIONONAPOLYS OITB will offer the following solutions:

› **PILOT LINES**: Cutting edge technology (see Figure 2) upgraded at TRL 7 aligned to the market demand, to mitigate the risks associated to integration of new technologies into marketable products.

› **PRIMARY RAW MATERIALS FROM DIFFERENT FEEDSTOCKS**: BIONANOPOLYS will use the most relevant feedstock in Europe to obtain bio-based nano-enabled materials and will ensure its availability during all year throughout a management and a logistic optimization of the most adequate biomass in each moment. **Primary sources of feedstock EUROPE**[1] supposes **1 Bt.** of agricultural crops (61.4%), grazed biomass (11.7%), forestry (26.6%), fisheries and Aquaculture (0.3%), **secondary 0.2 Bt**., additionally, from food waste[2], Europe generates around 89 Mt/year of food waste. 75% of the food waste is sent for incineration or landfilling. BIONANOPOLYS will use high lignocellulosic (mainly with high cellulose content) feedstocks for the production of cellulose nanofibers (CNF), Cellulose nanocrystals (CNC), Nanolignin (NL), and metallic nanoparticles (NP), and high sugar content feedstocks for the production of building blocks, organic acids, PHA and active compounds for the production of nanocapsules. A deep study of the biomass is going to be undertaken in order to select the most suitable final biomass sources within agricultural/forestry wastes, annual renewable plant, and food by-product and wastes. **HIGHLY DEMANDED APPLICATIONS:** BIONANOPOLYS will facilitate the market penetration of nano-enabled biobased materials by targeting the most demanded applications: packaging, textiles and consumer goods - which in total represent more than 50% of the total biobased material market share. Considering Bio-Based Chemicals , By type, bioalcohols

[1] http://bio-based.eu/reports/

[2] https://www.eu-fusions.org/phocadownload/Publications/Estimates%20of%20European%20food%20waste%20levels.pdf

accounted for the largest share of 41.7% by value in 2018 (24.653M$), followed by food and beverage (26%), bioplastics (11%), biolubricants (5%), biosurfactants (4%), biosolvents (3%) and others (10%)[3]. The Bio-Based Materials Market by Applications for the period 201-2026 (valued at EUR 10,000 million in 2017 and an expected revenue of EUR 85,150 million by end of 2026), growing at a CAGR of 27.1% between 2018 and 2026. According to our forecast (section 2), BIONANOPOLYS will be sustainable by capturing of the Serviceable Obtainable Market between 3.76% the first year and 5.67% the year 5 (with a cumulative revenue of €146M).

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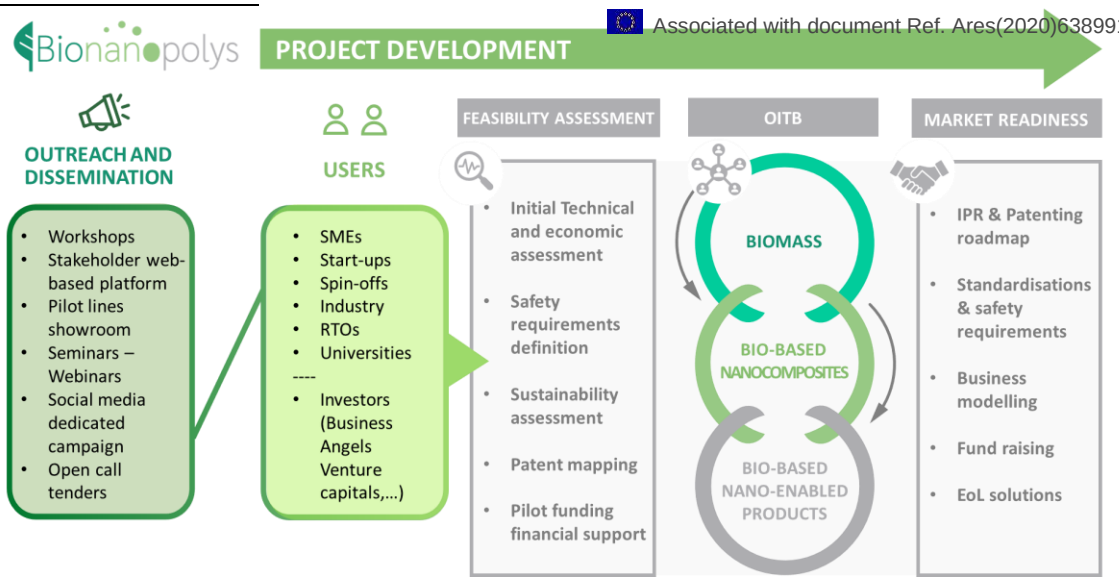

Figure 1. Overall concept of BIONANOPOLYS Test Bed.

The overall concept of BIONANOPOLYS Test Bed is shown in Figure 1. In addition to the network of pilot plants, BIONANOPOLYS OTIB will offer to the industry a wide variety of services for the market uptake of new bio-based nano-enabled products, such as safety protocols for bio-based nano-enabled materials, training for staff specialization, standardization, business modelling, access to follow-on finance or IPR protection as a crucial mean of ensuring the capitalization on the investments made by our stakeholders and other investors.

[3] https://www.marketresearchfuture.com/reports/bio-based-chemicals-market-5706

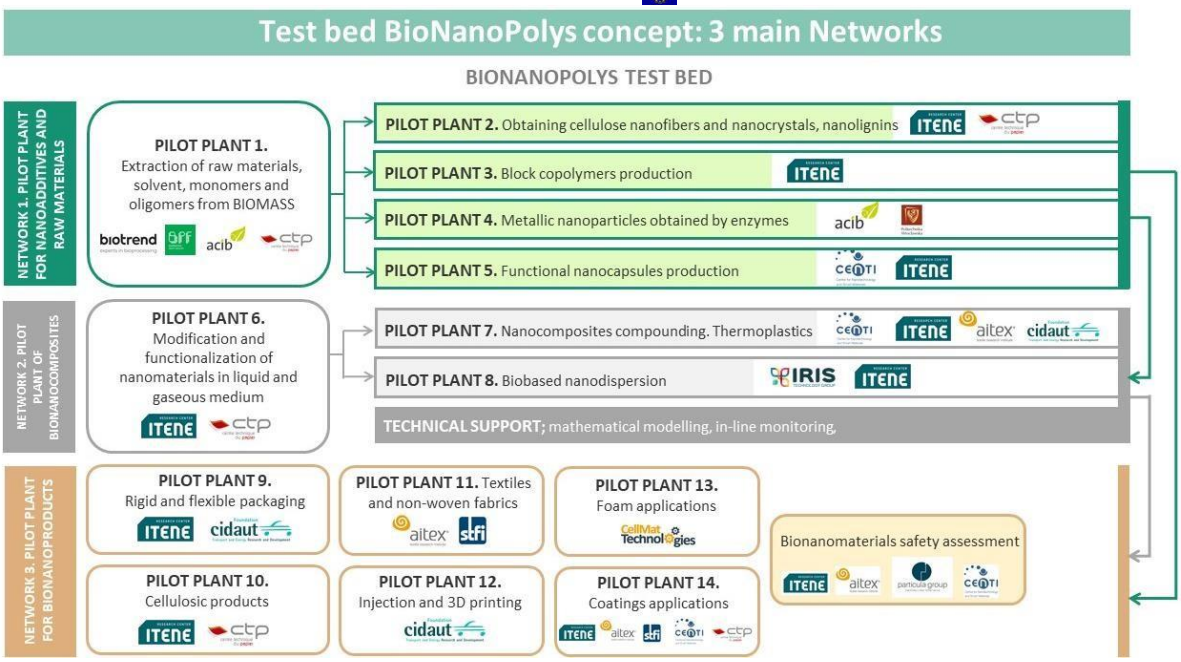

Figure 2. Pilot plants involved in BIONANOPOLYS Test Bed.

In order to push the introduction of high added value biobased materials to the industrial sector BIONANOPOLYS Test Bed will engage different pilot lines covering from the isolation of valuable compounds from BIOMASS, to the production of nanoparticles, the dispersion of this nanoparticles in different substrates, and targeting the final products covering different industrial applications. We will be organised in 3 nets and 14 pilot plants (see Figure 2).

The first network involves 5 pilot plants (PP). The first PP will work with the pre-treatment of Biomass, and isolation of valuable compounds in order to obtain our biobased nanoparticles, cellulose nanofibers (CNF), Cellulose nanocrystals (CNC), Nanolignin (NL), metallic nanoparticles (NP), building blocks (BB), block copolymers, nanocapsules (NC). The second Network will use these nanomaterials to obtain nanocomposites; in a first step surface modification or functionalization could be required to get a good dispersion in the final substrate. Then dispersion of nanoparticles will be performed through compounding, or nanodispersions. Finally, in network 3 the different Bionanoparticles, and bionanocomposites will be used to obtain different products that will cover a wide range of industrial applications.

1.1 Objectives

Overall Objective: Create a network of pilot plant and complementary services to speed up the introduction of biobased nano-enabled materials into the market through a Single-Entry Point (hereinafter SEP).
a) Create a strong platform to support companies into overcome any type of barrier to introduce biobased nanoenabled materials into the market; technical, legal, regulatory, safety, economic and financial. b) Penetrate the industrial sector reaching at least new 200 SMEs[4] around Europe by deeply studying the requirements in each industrial sector. c) Focus the technological pilot plant to market necessities and implementing the feedback from different actors. d) Maximise the impact of the OITB and the excellence by cooperating with other Test Beds and European bodies.

According to the specific risks above mentioned, the main objectives of BIONANOPOLYS will be the following:

Objective 1. Upgrade and fine tuning of 14 pilot lines across the entire BIONANOPOLYS value chain.

[4] As it is stated in section 2.1 impact, current clients from BIONANOPOLYS consortia in this sector is estimated at 600.

The first objective is to upgrade the 14 pilot plants (see Figure 2) to be ready to be opened to the European SMEs, implementing important technological updates to push our companies to use innovative product and process, specifically:

- 5 pilot plant focus on the development of bionanomaterials from biomass (High added value nanomaterials)
- 3 pilot plants to develop bionanocomposites (ensuring safety and maximizing NP dispersion)
- 6 pilot lines to manufacture bio-based nanoproducts from them (reaching a wide range of applications in different sector)

Detailed objectives are presented following:

e) Upgrade a pilot plant for the treatment of BIOMASS in order to obtain the required raw materials that are the basis for the development of the Nanomaterials (BPF, BIOTREND, ACIB, CTP) - PP1.

f) Upgrade four pilot lines to obtain biobased nanomaterials: cellulose nanofibers and nanocrystals (CNF, CNC), nanolignins (ITENE, CTP) – PP2, block-copolymers (ITENE) – PP3, metallic nanoparticles (PWR and ACIB) – PP4, and nanocapsules (CENTI) – PP5, in terms of yields, quality of the final products and energy consumption. Green chemistry will be applied to all the processes to reduce the environmental impact of the processes.

g) Upgrade three pilot lines to optimize the surface modification to confer good compatibility with the different matrices through green chemistry liquid–solid, gas–solid, and chromatogeny (ITENE, CENTI, CTP) – PP6, and the dispersion of biobased nanomaterials in polymeric matrices (bionanocomposites) (ITENE, CENTI, AITEX, CIDAUT) – PP7; compounding (melt extrusion), or suspensions (liquid dispersions) (ITENE, AITEX, CIDAUT) – PP8. In – line monitoring and fast characterization techniques will allow us to reduce trials time from days to minutes and increase product quality, optimizing industrial parameters.

h) Create an internal bureau to assure quality and safety of the different process and nanomaterials developed; in–line monitoring (IRIS, CEA), fast characterization (ITENE, AITEX, CIDAUT), modelling (IMT), nanosafety (ITENE), and food applications safety according to EFSA and FDA) (ITENE). All the required tasks will be afforded inside the test bed, so reducing time and barrier to market penetration.

i) Upgrade 6 pilot lines for manufacturing nano-enabled products: flexible and rigid packaging (ITENE, CIDAUT)- PP9, cellulosic products (ITENE, CTP)- PP10, coatings (ITENE, CTP, AITEX, CENTI) – PP11, textile and nonwoven fabrics (AITEX, STFI) - PP12, foamed (CELLMAT) – PP13, and injected and 3D printed (ITENE, CIDAUT)
- PP14 bionanoproducts. This will allow us to reach a wide range of applications in different industrial sector.

j) Demonstrate the technical performance of the pilot plants within more than **twenty case studies involving seven key industrial players**, who will manufacture the innovative products developed: Novamont (reinforced polymers for packaging), Ambrosialab (cosmetics and functional ingredients), Textisol (textile and non-woven), DANIPACK (Flexible food packaging), DS Smith (packaging paper applications), CELLMAT (foamed packaging products and light-weighting structural parts) and Logoplaste (rigid packaging), HTSA (agricultural synthetic fibres).

Quantification of objectives is performed in the concept section by defining the improvements to be carried out in each pilot plant in order to reach 15% improved industrial process parameters and 20% faster verification of materials performance and productivity.

Objective 2: Implement a transversal technology bureau to suppor

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pilot plant and end users by reducing technical risk, time and barriers to the market penetration.

k) The project will assure responsible and safe use of bionanomaterials along the entire value chain of industrial innovation processes, from occupational and environmental exposure to consumer safety and regulation aspects. Safety by Design (SbD) approaches will be applied, together with the design and selections of tested risk management measures to control the exposure and reduce release of nanomaterials to the environment. To achieve this objective, a close cooperation with the EU Nanosafety cluster (NSC) community, as well as international networks is envisaged (ITENE).

l) Evaluate and optimize the environmental, economic, social effects of the test bed across the full supply chain (including the design, manufacture, and use of the project results) though Life Cycle Assessment (LCA), Life Cycle Costing (LCC), Social Life Cycle Assessment (S-LCA) and eco-efficiency analyses (Particula). An additional end-of-life assessment will ensure recycling, composting or biodegradability of the manufactured products (ITENE, CIDAUT, AITEX, CTP, BPF, CENTI and CELLMAT).

m) Develop safety protocols for food contact, and cosmetic applications & regulatory assessment (e.g. migration tests to ensure Food contact: overall (EN 1186) and specific (EN 13130), European Food Safety Agency (EFSA) dossier services when required) (ITENE).

Objective 3: Launch and Implement Innovation management services for facilitating the commercial exploitation of the activities done by clients and users of BIONANOPOLYS.

n) Implement a funding support service to SMEs, with the support of EBN and EBAN engaging 100 companies introduced to the investor community and 150 supported projects.

o) Develop strategies transferable to the users of the IOTB on legal issues and IPR (G&S).

p) Implementation of regulatory and safety standards needed for market deployment (ITENE, G&S).

q) Develop strategies to access to investment capital instruments, with the support of EBN and EBAN.

r) Provide mentoring to companies' staff to improve their skills on the technology uptakes (EBN, EBAN).

s) Implement an open call to evaluate customer satisfaction and improve weak points.

Objective 4: Launch an accessible and economically sustainable Test Bed during and after the project

t) Define the legal entity structure by M22 (AXIA, covered by G&S for legal and compliance, and rest of partners except end users).

u) Create a user-friendly web-based single-entry point (SEP) in all official languages of participating countries: English, Spanish, French, Portuguese, Romanian, Dutch, Croatian, German, Flemish, Polish and Italian.

v) Create and update IP models addressing ownership, protection and legal and compliance, to maximise the potential of collaboration- led by the law firm G&S.

w) Define a Business model to exploit BIONANOPOLYS (AXIA, EBAN, EBN, G&S) to obtain a turnover of over 48 M€ (four times EU funding) within five years after project ends.

x) Implement a dissemination and communication plan at all level (regional, national, European) to reach as much as possible of potential consumers approached from different sectors (ACIB).

y) Bring over at least 5 early adopters though an open call, carry out 20 feasibility assessment and reach out at least 400 stakeholders (50 per user case) through outreach activities, including a digital platform for stakeholder engagement (All partners).

1.2 Relation to the work programme

Novel approaches to produce nano-enabled bio-based materials with advanced properties and functionalities. Demonstration in an industrial environment the conversion of different types of feedstock and bio resources.

BIONANOPOLYS will guarantee the sustainability of the feedstock used to obtain different nano enabled biobased materials (NEBM) its supply and logistics (UGENT). To maximise the use of different biomass sources from Europe, the OITB will use the expertise developed in previous projects (see Table 1), and, additionally cooperation with other European projects will be considered. BPF, BIOTREND, ACIB and CTP will valorise this biomass by extracting different raw materials, needed to produce NEBM. Lignocellulosic based biomass, such as agricultural crops and grazed biomass and Food waste, coming from the food industry and also from the agricultural wastes, will be the mainly feedstocks selected for the production of the nano enabled bio-based materials. Additionally, certain organisms, from higher plants, algae, and microorganisms from various kingdoms will be used to produce metallic nanoparticles.

Scope:

Upgrading existing pilot lines, and make available, their services for the design, development, testing, regulatory (including safety) and environmental assessment, and upscaling of specific material compositions;
Besides the upgrading objectives defined in 1.3 Concept section, BIONANOPOLYS project development starts in eco-designing the product itself and ends up with the end of life of the product, evaluating and prioritizing recycling, biodegradability and compostability based on the applications. In between, the following activities will be available to avoid any hurdle to reach the market: safety and sustainability assessment, business development, access to follow-on finance, feasibility study, market strategy and standardisation.
The focus is on new nano-enabled bio-based materials' functions, features, capabilities, robustness and properties, processing techniques and optimisation of process parameters,
RTO's involved in the project have a broad experience in the valorisation of biomass to obtain different materials. Specifically, NEBM will be developed to achieve multifunctional properties. CNF will be used for improvement of mechanical and barrier properties, modified natural clays will be used to improve mechanical, thermal and barrier properties (gases and grease), nanocapsules will be developed to produce active effect (antioxidant, antimicrobial, self-healing, aroma or ingredient release, etc.), metallic NP will confer to the materials antimicrobial effect, and block copolymers will be synthesized as tailor-made nanoreinforcements for different polymer matrices. The isolation and modification of NEB will lead to new materials as are tailored to purpose, being more specific than nowadays available. The proper dispersion of the NP in the different substrates will be follow by in line monitoring when possible, and with fast-on-line characterization protocols in the other cases. This will ensure the optimization and robustness of the process parameters and final properties of nanoproducts.
Proposals should assess regulatory, safety, economic and technical barriers and should develop and validate tools to enable in-line quality control processes;
BIONANOPOLYS will work on the market validation, developing a market uptake strategy (EBN) and IPR analysis (G&S Law), identifying fund raising opportunities (EBAN). The OITB will offer safety assessment services (food safety and specific protocols for environmental, health and safety (EHS) assessment (ITENE)), besides we will look for synergies with other OITB to characterize the NPs developed. IRIS and CEA will set up and calibrate in-line quality systems that will allow measuring the NPs dispersion in real time during melt processing, and the coating application, respectively. The system will be integrated in the compounding and coating pilot plant, in order to assure quality dispersion of NP. Due to its high aspect ratio and dimensions, even low feeding ratio provides a remarkable number of particles that need to be properly and evenly distributed. It must be remarked that nanoparticle impact on final properties rely on both a proper compatibility with matrices and a dispersion on the matrices, preventing its agglomeration. Therefore, proper dispersion of the NP in the different substrates will be follow by in – line monitoring when possible, and with fast-on-line characterization protocols in other cases. This will also ensure the optimization and robustness of the process parameters and final properties of nanoproducts.
Open access at fair conditions and cost as well as outreach and dissemination across Europe, based on a distinct methodology;

BIONANOPOLYS has participants from 12 central, Western, Eastern and southern European countries (SP, UK, FR, PT, NL, AT, DE, RO, PL, HR, BE and IT). The project will promote business opportunities and collaboration between its members and industry, public organisations and research and educational institutions, based on an open access principle. The BIONANOPOLYS platform will be built on transparency, social networks and knowledge exchange. Besides, the pricing will be clearly defined for BIONANOPOLYS users to assure the fair conditions of access. As defined in Figure 1, a variety of outreach and dissemination activities will be done to get early adopters.
Materials should be demonstrated in relevant industrial environments;
To guarantee the quality of BIONANOPOLYS' technologies, user cases will be implemented at TRL 7 during the project, according to the needs of our validator partners, the SMEs: CELLMAT (SP), AMBROSIALAB (IT), Textisol (SP), HTSM (SP) and large industries: LOGOPLASTE (PT), NOVAMONT (IT), DSSMITH (NL), DANIPACK (PT) . These use cases will be tested in an industrial environment at market conditions. Additionally, the test bed will be opened to at least five additional external users through the open call in order they can validate their products.

1.3 Concept and methodology

The main purpose of this OITB is to open cutting-edge pilot lines, in particular to SMEs, to develop or design specific nano enhanced biomaterials, aiming to provide required properties or performance (barrier, active, mechanical, thermal, properties, etc). To reach out those future users, different events will be held, consisting of in-person

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Workshops, webinars, pilot lines showroom, social media dedicated campaign, open call tenders for early adopters and outreach videos/animations as well as a web-based platform. BIONANOPOLYS will engage end-users through these targeted outreach activities to promote the capacities of the OITB in developing tailor-made projects ready for the market (see Figure 1).

BIONANOPOLYS PROJECT DEVELOPEMENT

Any user of the OITB will be able to access through the SEP at any point of the project to a specific development stage within the project layout or concept, trying to cover its needs within BIONANOPOLYS capabilities (Figure 1).

1. OUTREACH ACTIVITIES

BIONANOPOLYS targets the bio-based and nanomaterials sector to communicate and disseminate the capacities of our technology and how we override any risk for the market deployment of new innovations on the field. To do so, BIONANOPOLYS will implement several dedicated activities to create a brand as it is indicated above, in order to become a reference on the sector and create a brand distinguishable to facilitate the market penetration.

2. FEASIBILITY ASSESSMENT

At the SEP the potential users will find a complete roadmap that will pave the way for the market readiness. The roadmap includes a menu of services to accelerate the market exploitation. Starting with a preliminary market assessment and LCA, identification of the best End-of-Life (EoL) solution and an initial technical and economic feasibility of the project and long-term sustainability (including safety and regulatory issues).

3. OITB

BIONANOPOLYS concept is formed by three main networks (see Figure 2). The **network 1** regarding pilot plants for **nanoadditives and raw materials**. This network will oversee the selection and logistics of biomass, its extraction of raw materials and production of different nano additives by the treatment and/or reaction of these raw materials.

Pilot plant 1

Extraction and isolation of raw materials, solvent, monomers and oligomers from BIOMASS.

TRL4 to **TRL7**

Enzymatic pre-treatments will be carried out in high solid load (20 -25 %) in order to produce a solution with high sugar content (BPF).

Extraction and isolation of raw materials, solvent, monomers and oligomers from BIOMASS.

Cooking and bleaching pilot plant will be upgrade for the extraction of cellulose from annual plants by the process optimization in terms of chemicals, temperature, tim e, among other (CTP).

Multipurpose bioprocessing pilot plant with stainless steel, steam -sterilizable fermenters of 10L, 50L and 200L working volume and diverse pilot scale DSP equipment. Multipurpose **bioprocessing pilot plant**; with steam-sterilizable fermenters of 10L, 50L and 200L and diverse pilot scale DSP equipment. The plant will be upgraded with advanced process monitoring capabilities (fermentation and DSP) and integrated control software to enable monitoring and control at process level instead of unit operations level. Implementation of Industry 4.0 features, including remote sensing and control. Production al lactic acid and succinic acid with a conversion rate >90%. High PHAs production yields (> 5kgs/batch). (Biotrend).

Obtaining cellulose nanofibers and nanocrystals, nanolignins.

Pilot plant 2

Obtaining cellulose nanofibers and nanocrystals, nanolignins. **TRL4** to **TRL7**

Implementation of a continuous production process and new grinding stones and therefor increase production capacity from 1kg/day to 2kg/day dry CNF (ITENE).

Twin screw extrusion process will be upscaled to obtain CNF with a high consistency, from 20% to 30% solid content, and reach a capacity from 2 kg/h up to 20 kg/h. This upscaling will be performed by adapting the process parameters and the screw profile (CTP).

Obtaining block copolymers.

Pilot plant 3

Obtaining block copolymers. **TRL4** to **TRL7**

Upgrade 3 - 10 L polymerization reactor with polycondensation system and higher toque stirring system to ensure a proper reaction. By being able to use this reactor, production capacity will increase from 0,5kg/day to a maximum of 7 kg/day. A fast characterization method will be also implemented to assess the optimization of final products.

Pilot plant 4

Metallic nanoparticles obtained by enzymes.

TRL4 to **TRL7**

Using biomass from terrestrial annual plants, with phytoremediation potential, they will be used for the production of Ag and other metals nanoparticles in conversion ratios above 80%.

Pilot plant 5

Functional nanocapsules.

TRL4 to **TRL7**

Upgrade of the **nanoemulsion production process** by the implementation a continuous process to wash the nanomaterials from the reaction medium, allowing exchange in solvent or production of nanomaterials in powder format (CENTI).

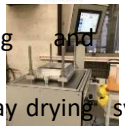
Upgrade of the **nanocapsules drying process** by the installation of a new inlet nozzle with smaller diameter with aim to improve and control/reduceOptimization of the process in terms of inlet and outlet temperature, air flow, yields, among others (ITENE). nanocapsule particle size and adapt other parameters to reduce particle size distribution.

The **network 2** regarding pilot plants for production of **bionanocomposites** and the pre-treatment needed to process the bionanomaterials that will be introduced in this bionanocomposites.

Pilot plant 6

Starting from NANOMATERIALS

Modification and functionalization of nanomaterials in liquid and gaseous medium. TRL4 to **TRL7**



Optimise surface modification method (nanomaterials and substrates) applying green chemistry and biobased raw materials (ITENE and CTP), reducing solvents at least by 20%. Increase production rate of modified nanomaterials from 2 to 4 kg/day by incorporati ng new feeding and dispersing systems enabling higher solids load and leading to higher reaction yield (ITENE).

Increase content of solids from 1 to 2% of nanocellulose and from 10 to 20% of nanoclays in spray drying system (ITENE) by improving the stirring system during the feeding of the suspensions.

Pilot plant 7

Nanocomposite compounding. Thermoplastics.

TRL4 to **TRL7**

Increase of compounding productivity from 20 to 27 kg/hour (CIDAUT)

In-line monitoring system (ITENE) and tailored configuration of the compounding technology to ensure a proper dispersion of nanomaterials to obtain high performance nanocomposites in terms of mechanical, barrier and active properties (ITENE, CIDAUT and CENTI), thus reducing trials time by at least 30 %.

High accuracy (0.01-5 mL/min) of nanoadditives feeding in liquid state and with a constant flow achieving more homogeneous materials in **reactive extrusion process** (AITEX).

Pilot plant 8

Biobased nanodispersion. TRL4 to **TRL7**

Biobased nanodispersion.

Increase of polymeric **nanodispersions production** in solvent based systems from 20 to 100 liter/batch (ITENE).

The plant will be upgraded with a new reactor of 100 liter, with an emulsifier and a ultrasound tip to increase dispersibility and stability of suspensions, with this, it is intended to increase the performance of the

modification to improve at least 20% in industrial productivity, reducing production time and use of solvents. Pilot scale reactors will be upgraded with a new stirring system (homogenizer for higher velocities) for dispersion of nanomaterials (CENTI) in polymeric resins. Increase from 1 to 1.3 kg/day of nanocomposites

Finally, the produced by **network 3in situ polymerization** regarding pilot plants to obtain by the adaptation of the rectors to new larger polymerization vessels**bionanoproducts** from the different bionanocomposites of Network 2(CIDAUT)..

Associated with document Ref. Ares(2020)6389918 - 05/11/2020

Pilot plant 9 Starting from BIONANOCOMPOSITES

Packaging. Rigid and flexible applications

TRL5 to **TRL7**

Cast Extrusion for multilayer structures (ITENE): intermediate layer will be used to add additive compound or nano-additives to control release or avoid direct contact with the food or cosmetic product. For **monolayer structures (CIDAUT)**, the plant will be upgraded to obtain thin films based on biopolymer matrices and different types of nanomaterials, which will improve the properties and functionalities of the polymer matrix to offer services for different biobased polymers.

Thermoforming (ITENE): process with the aim of evaluating the properties of the bionanocomposites in an actual packaging format.sheets of multilayer and monolayer structures can be converted in a rigid packaging by this

Cellulose based applications

Pilot plant 10

Cellulose based applications

TRL5 to **TRL7**

Cellulose based applications pilot line will be upgraded to move from batch process CNF-polymer film production to continuous pilot line to be able to increase the strength of the wet mat which will be obtained by the use of nanofilaments and optimize the drying of the wet mat onto a hydrophobic filter paper. The obtained cellulose nanofibers webs/reels will be treated with chromatogeny technology to bring water barrier properties (CTP).

Textiles and non-woven applications

Pilot plant 11

Textiles and non-woven applications

TRL5 to **TRL7**

Extrusion nonwoven line will be upgraded with a multirow nozzle unit (8 rows of capillaries, 1800 active capillaries in total) with capillary diameters of ≤ 0,3 mm.

This will lead to finer filament diameters and a higher efficiency of the process in terms of material throughput (50 kg per hour) compared to the single row system. Furthermore, the plant will be upgraded with a hydrocharging unit that will enable the surface modification of the filaments (electrostatic activity) during the **meltblown process** online. This will strengthen the application of the materials in filter sector (STFI).

Fibre nonwoven line (carding, needling, stitch-bonding) will be upgraded in terms of control unit and digital interfaces which will improve the efficiency of the production by: Control system with novel drive regulators, SPS control and data storage, drive systems with motors/gear motors for the carding system and servo drive for the crosslapper, quality

Injection and 3D printing.

insurance system for an online monitoring of the product quality, safety systems to protect workers and machinery, user interface with modern data visualization, management of recipes and remote data transfer, predictive maintenance (STFI).

Nanobubbles system will be upgraded to provide the option to work under nitrogen atmosphere could solve the oxidizing problems observed in natural and bio-based chemicals for finishing and dyeing. This upgrading will extend the range of raw materials used for finishing and dyeing garments, providing new colour effects and a high-valuable sustainable finishing/dyeing process to the garment and fashion industries (AITEX).

Pilot plant 12

Injection and 3D printing.

TRL5 to **TRL7**

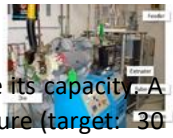
Injection of Rigid packaging the improvement will be to implement new moulds for cosmetic applications and preforms (bottle and jar format) (ITENE), increasing market applications. **3D printing (FFF)** consists of two different devices: filament maker (extruder machine, path cooling and filament winding) and two homemade printers. Printers will be modified (or redesigned and manufactured) for ensuring the printability (speed, dimensional accuracy and surface appearance) of different thermoplastic bio-based nanocomposites (CIDAUT).

Pilot plant 13

Foam applications

TRL5 to **TRL7**

Extrusion foaming pilot plant will be upgraded with new equipment to work with bionanocomposites/bionanofillers such as dust-free gravimetric dosifiers, a new gear pump, engine power increment, side feeding extruders, vaccum system attached to the barrel to release steam generated, new flat die to increase the size of products produced. and in/on-line measurements of viscosity, cell size, thickness and density (CELLMAT).



Bead foaming pilot plant (micropelletizing + expansion + thermoforming): will be upgraded to increase its capacity. A new vessel will be acquired with a higher capacity (5 l/batch) and a higher thickness to support more pressure (target: 30 MPa). A new electric valve to increase pressure drop rate will be also installed. The micropelletizing system (twin screw extruder + pelletizing) will be upgraded to install in-on line rheology measurements and a more robust cutting machine. A steam-chest mould for laboratory will be acquired to produced thermoformed boards from biobased beads (10 x 10 cm) that will be used to check mechanical properties improvements due to the reinforcement of biofillers.

Pilot plant 14

TRL5 **Coatings applications** to **TRL7**

The pilot scale roll-to-roll will be upgraded with ultrasonic spray-coating with a velocity of 2 m/min. The plant would be upgraded with an enabled and continuous process that integrates thermal and/or UV-light cure. With a final capacity of spray with multiple nozzles and nozzles' geometries at 2-10 m/min speed with integrated cure stage. Also, a pilot scale **roll-to-roll slot die coater** with a 2 L pressure vessel. The plant will be upgraded in terms of the pressure vessel and its compatibility with different materials. A final capacity to apply 15L of nanomaterials as nanocoatings as well as the increase of the number of nanomaterials that can be used (at least, more 3). To improve the production process, the quality of the nanofibers and to extend the range of bio-nanomaterials (e.g. nanofibers for biomedical applications) to be obtained (CENTI)

Coatings pilot line (flexo and gravure): In this pilot plant coatings are applied on polymeric and cellulosic substrates. Coating thickness and homogeneity is crucial to reach superior barrier properties. This pilot plant will be upgraded by in line monitoring system for thin layers measurements and homogeneity assessment in a roll-to-roll machine (ITENE), reducing trials time from days to minutes and increasing product quality, optimizing industrial parameters.

Electrospinning line will be upgraded to improve the production process, the quality of the nanofibers and to extend the range of bio-nanomaterials (e.g. nanofibers for biomedical applications) to be obtained by integration/modification of specific parts and devices of the electrospinning system (AITEX).

3D spray-coatings line 3D cellulose-based objects will be spray-coated after their 3D shaping the pilot line will be sprayed to impart additional performances such as oxygen and grease barrier properties. The environment of the 3D spray-coating pilot will be adapted (pumps & tubes) to be able to treat CNFsurface will be adapted to ensure homogeneous coverage of the 3D-biopolymers formulations. Trajectory, flows, speed to -cellulose based packaging (CTP).

Additionally, BIONANOPOLYS test bed will cover other technological services closely linked to the introduction of new products into the market: 1- Create an internal bureau to assure quality and safety of the different process and nanomaterials developed (including nanosafety, REACH, and Food applications assessment (EFSA) . 2- In-line monitoring and fast characterization. 3- End- of -life (recyclability, and biodegradability) evaluation.

4. READINESS FOR THE MARKET

BIONANOPOLYS integrates a business acceleration service. These include support to access to follow-on finance with the support of EBN and EBAN through brokerage activities and fund-raising events. The OITB will offer training and assistance to apply for further EU risk finance, or R&D public funding at local, regional or European level. Furthermore, the user of BIONANOPOLYS will find a range of other support activities and such as IPR protection, Regulatory and Standardisation or Business modelling with the support of G&S Law.

Related Projects (Table 1)

BIONANOPOLYS will establish relationships with ongoing and future projects from this topic and related calls (BBI, SPIRE, NMBP) to achieve common goals settled by the biobased industry, and supporting digitising and transforming European Industry, and maximize the exploitation of previous results. All steps have been covered: biomass conversion, bionanocomposite development and bioproducts, regulations/safety and other OITBS.

*Table 1: interactions with other projects involving production of nano-enabl bio*Associated with document Ref. Ares(2020)6389918 - 05/11/2020*-based materials (acronyms linked to its website; coordinators of project underlined)*

Project and partner	Result from the project interacting with BIONANOPIOLYS
BIOCATPOLYMERS BIOTEC-17 BPF	bio-chemical catalytic cascade conversion of residual biomass to high quality biopolymers: pretreatment of biomass, biological fermentation, selective catalytic conversion to monomers, purification, and polymerization.
AGRIMAX BBI-15 (IRIS, ITENE, UGENT)	Pilot lines of valorisation of agro-waste: biorefinery processing technologies: NIR for monitor biomass composition, extraction, and enhancement of nanocellulose fibres, biopolymers, and active compounds, and validation in packaging applications.
ENZYCLE BBI-19 ITENE, ACIB	Development of pre-treatment processes to enhance enzymatic activities. Development of enzymes and advanced bioconversion processes. Purification of building blocks. Polymerization of 2nd generation building blocks.
COMPATIBIO Polynat-ANR project CTP	Bio-based Thermoplastic Composites Compatibilized by Chromatogenic Chemistry
PROTECT NMBP 2016 (AITEX, CENTI)	Pre-commercial lines for production of surface nanostructured antimicrobial and anti-biofilm textiles and medical devices.
OPTINANOPRO NMBP-15 (IRIS,ITENE)	Optimized nano enabled materials for packaging and automotive, composition monitoring techniques
REINVENT BBI-17 CeNTI	Up-scale technologies to produce bio-based fibre and foam composites applied in automotive.

PULPACKTION BBI-15 (IRIS, ITENE, NOVA)	Pilot lines for cellulose-based tailored-to-purpose food and non-food packaging solutions: multilayer films, composites and coatings. Wet-moulding, thermoplastic starch, PLA, MFC reinforcements.
POLYBIOSKIN BBI 2016 (IRIS, UGENT)	Skin-contact biopolymer-based products for personal care, Cosmetic and biomedical industries, including films and textiles.
SCALIBUR SC2-2018 (ITENE, IRIS)	Development of OFMSW pre-treatment processes through enzymatic processes and subsequent bioconversion to produce 1,4-BDO. Polymerization of new polyesters and validation of new materials.
DEEP PURPLE BBI-2018 (ITENE).	Development of cellulosic biomass pre-treatment and conditioning processes. Development and optimisation of enzymatic hydrolysis processes and downstream fermentation processes to produce building blocks for the synthesis of new materials
NANOUPTAKE NMBP 2019 (ITENE)	Computing infrastructure for testing and implementation of safe-by-design approaches in shared nanotechnology sectors: packaging, automotive, pharma.
LEE BED: OITB 2018. (ITENE, AXIA)	NFC & nanocrystals from BIONANOPOLYS will be sent to LEE BED to study its applicability on printed electronics (formulation of conductive inks). Lessons learned from AXIA in innovation management of OITB and SEP will be put into practice.
SKHINCAPS (CENTI)	A research project aiming to develop customised and safe nanocapsules to deliver novel products for skin healthcare applications, with increased efficiency and cost benefits, leading to ground-breaking innovations on the actual products.
SAP4MA (CENTI)	Development of a printed active device using nanomaterials to produce smart packaging solutions for margarine targeting the extension of the shelf-life of the packed margarine through the controlled release of active natural antioxidant agents at the onset of the food degradation process.
NENU2PHAR BBI-19 (CEA, ITENE)	Development of PHAs from algae biomass, obtaining a high purity product by an optimized process, at competitive price. To meet the requirements of current packaging materials, these PHAs will be reinforced with block-copolymers, microfibrillated cellulose and nanocrystals and starch nanocrystals.
ASINA (CENTI)	Development of concepts to ensure safety of nano-enabled products through design. ASINA will formulate and design hypothesis and make design decisions by applying a data-driven approach and methodology (the "ASINA-SMM").
Alliances with OITBs: i-TRIBOMAT, REALNANO, CHALLENGES, MMAMA and OYSTER	Shared promotion between stakeholders and clients, visits for study alliances to exploit and share current pilot plants and infrastructure. E.g. characterization services, tribological properties: wear, friction, in line monitoring of nanocoatings, coating/printing characterization.

b) Methodology

LEGAL ENTITY and SINGLE-ENTRY POINT (SEP) : the consortium will assess **by M22** a Legal Entity structures as the main exploitation pathway for the OITB SEP. A user-friendly webtool will be developed to allow the easy access to the services of BIONANOPOLYS. It will be available in different languages. The SEP will include the capabilities of the Pilot Lines, the complementary Business Accelerator Services and the information about outreach activities and open calls. **WP 2 (lead by AXIA)** will ensure an optimal management of the legal entity and SEP.

PILOT PLANTS : BIONANOPOLYS will upgrade their pilot plants **by M32** as explained in Figure 2**¡Error! No se encuentra el origen de la referencia.**. **WP3-5 (lead by ITENE, CTP and CENTI)** have been define for upgrading respectively the plants from each network. **WP6 (CEA)** will complement them with monitoring and modelling tool. Pilot plants will be tested by end users in **WP7 (AITEX)** from M24 to M42 and by external users through the open call in **WP10 (ITENE)** from M28 to M48.

COMPLEMENTARY SERVICES : The safety and sustainability assessment, business development (Follow-on finance, R&D projects, public funding, feasibility study, market strategy) are complementary services offered by the platform. **WP8 (PARTICULA)** and **WP9 (EBN)** will ensure its applicability in the OITB and will define a portfolio of services for end users.

OUTREACH and EXPLOITATION: It is expected to obtain an additional turnover of at least 4 times the requested EU funding, within 5 years of the end of the grant, which is more than EUR 48 Million. To go beyond this additional turnover and guarantee the sustainability of BIONANOPOLYS at least each pilot line will be available the 25% of its workability per year (working hour/year). A communication, dissemination and exploitation plan will ensure this objective **(WP11-ACIB).**

SYNERGIES : BIONANOPOLYS will implement collaborative activities with other OITB, such as MMAMA and OYSTER, that are offering services of characterisation. Different joint activities are planned (workshops, shared exploitation strategy). Furthermore, some members of the consortium such as ITENE, AXIA, CEA or IRIS are currently participating in ongoing OITBs projects, this will strength the collaboration and communication between Test beds and other projects (Table 1).

GENDER DIMENSION

The project believes in the methodology for Integrating sex and gender into engineering
g Past Innovation Practices; 2. Building the Design Team; 3. Obtaining User Input; 4. Evaluation
-step process will lead to new products, processes, or services and identify new markets and
s or meet the needs of a complex and diverse user group[5]. Besides, as the project uses include Designing Health & Biomedical Research concepts as well, for example, some
ffect human reproductive health, and this has been studied predominantly in men[6]. For
and Planning. This fou
a strong driver to replace formaldehyde, it is of concern during pregnancy because of
nanomaterials, it must
erinatal fetus**.** On the other hand**,** Women constitute 52% of the total European opulation but only
-employed and 30% of start-up entrepreneurs'[6] which means that 'female creativity
risk to the exposed 34.4% and
entrepreneurial potential -exploited source of economic growth and jobs that should be further
developed.')POLYS will follow the initiative led by the EIC Accelerator to promote gender
the BIONANOPOLYS OPEN call balance, thus will be supported by a communication campaign to ensure
submitted by women to have at least 25% of -led SMEs.

[5] http://genderedinnovations.stanford.edu/methods/innovation.html [6] Damstra et al., 2002.

[6] European Commission: https://ec.europa.eu/growth/smes/promoting-entrepreneurship/we-work-for/women_en

1.4 Ambition

BIONANOPOLYS will create for the first time an integrated platform of technologies, and scientific and technicians devoted to the nanotechnology based on bio based raw materials. Our team is specialized in nanotechnology in different sectors, with a high experience in European Projects, and Pilot Projects. The background acquired by partners will allow us to perform almost any project in the field of nanomaterials, and nanocomposites. Moreover, the range of application is enough broad in the OITB, and technical experts are capable to translate the technologies to other fields of applications. Our differential value is the high-quality pilot lines and the excellence in resources as well as the exploitation of market niches with bio-based products (see **¡Error! No se encuentra el origen de la referencia.**).

1.4.1 State-of-the-art and beyond: BIONANOPOLYS pilot lines.

1.4.1.1 Network 1. Nanoadditives and raw materials pilot plant.

Extraction of raw materials, solvents, monomers and oligomers from biomass.

Biomass waste presents a huge variability in its composition[7]. Depending on the source of the waste and effective decomposition of some of these components (cellulose, hemicellulose and pectin), simpler compounds can be obtained that are easily assimilated by microorganisms (sugars and amino acids). These compounds are considered as the limiting factor in the generation of by-products, since are key to define yield and process efficiency.

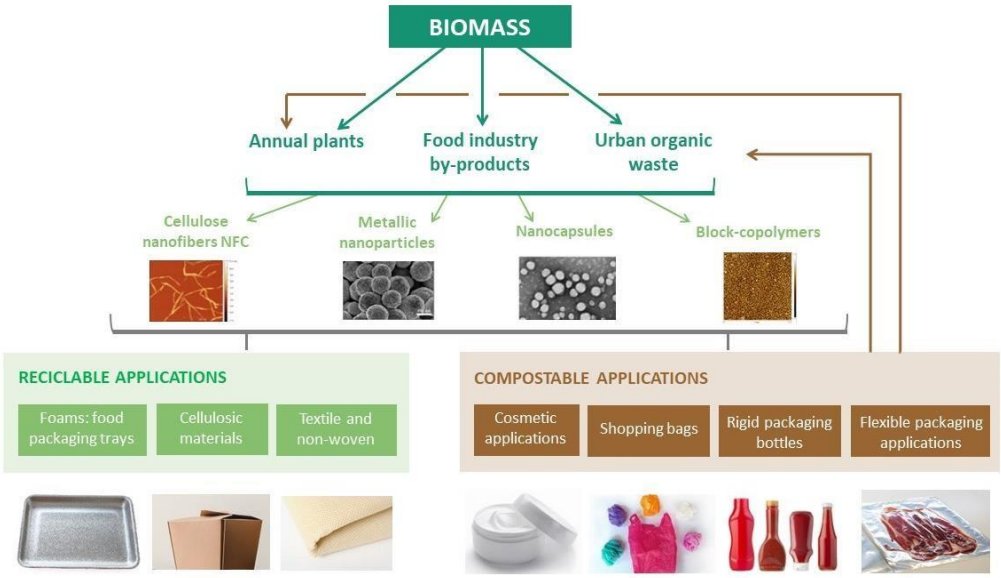


Figure 3.Products developed in BIONANOPOLYS Test Bed.

In BIONANOPOLYS Test bed different **biomass sources** will be used; lignocellulosic based biomass, such as agricultural crops (cotton fibres, hemp, flax, bamboo, sugarcane bagasse, etc.) and grazed biomass, with a high cellulose content will be focus on the extraction and isolation of cellulose to produce cellulose nanofibers (CNF), cellulose nanocrystals (CNC) and nanolignin (NL). Food waste sources will be also contemplated, mainly for synthetize building blocks, organic acids and PHA by biotechnological processes. Starchy and high sugar content feedstocks are the main substrate for the bioproduction of these raw materials. They can be produced from food sub-products, corn and other starchy plants using microbial fermentation technology. Fermentable sugars are generally a mixture of six carbon sugars (C6), such as glucose, and five carbon sugars (C5), such as xylose and arabinose. Other high added value compounds, such as active compounds with antimicrobial and antioxidant properties will be also isolate from the different types of treated biomass to produce active nanocapsules. Additionally, certain organisms, from higher plants, algae, and microorganisms from various kingdoms are capable of accumulating conjugated salts of various metals, that will be used to produce metallic nanoparticles.

[7] Ververis, C., et al. (2007). Cellulose, hemicelluloses, lignin and ash content of some organic materials and their suitability for use as paper pulp supplements. Bioresource technology, 98 2, 296-301.

Two main processes are going to carry out for the extraction of raw materials from biomass.

(1) Cooking and bleaching processes for the cellulose isolation from lignocellulosic sources. The extraction of cellulose is a well-known process within the paper industry. The cooking process consist in a thermal treatment of the feedstock in a basic medium with the aim of dissolve the lignin, hemicellulose and other component of the lignocellulosic sources. The slurry obtained is then treated in a bleaching process; treatments with peroxide, chlorite and others chemical have been applied along the years in the paper industry. This step removes the colour compounds of the pulp obtained in the cooking step.

(2) Bioprocesses include different units in which a near optimal environment is desired for microorganisms to grow, multiply, and produce a desired product.

The production of fermentable sugars from lignocellulosic waste from various sources is a well-known process within the state of the art. The hydrolysis process consists of several stages: liquefaction, pre-treatment processes and saccharification processes, starting mainly from the cellulose, hemicellulose and pectin present in the biomass.

For this reason, many **pre-treatment protocols** have been applied to this type of waste, with the aim of releasing the greatest amount of assimilable nutrients into the environment for downstream fermentation processes in charge of BPF. **For the production of building blocks and PHA fermentation from sugars is needed**, ACIB and Biotrend will optimized and scale up the fermentation processes. The first saccharification hydrolysis processes were carried out using acid (70% concentration) as a catalyst, at a temperature of 260°C, obtaining yields of 70%[9]. An alternative to this process is enzymatic saccharification (enzymatic hydrolysis), capable of reducing energy requirements, pressure, temperature and pH, as well as being more sustainable due to its low levels of contamination[10]. In the process of enzymatic saccharification of lignocellulosic biomass, a great variety of enzymes are used as catalysts to obtain glucose monomers, obtained industrially from fungi of the genera *Trichoderma, Aspergillus* and *Penicillum*[11], such as exo-cellulases, endo-cellulases and β-glucosidases. The selected enzymes vary depending on the nature of the starting material and on their hydrolytic capacity of the microorganisms of interest to be cultivated in the subsequent processes.

Companies like BASF showed interest in low-cost production of large quantities of sugars from lignocellulosic biomass (Renmatix in Pennsylvania, USA, process PLANTROSE®) [12]. Later in 2015, the French energy group, TOTAL S.A., formed a Joint Development Agreement (JDA) with Remantix to use PLANTROSE technology to carry out the extraction of second-generation sugars with the intention of producing high value-added compounds. Thus, in this line, in 2025, it has been estimated that biotechnological processes related to saccharification processes from biomass to obtain high added value products, will reach a market value of close to 1000 billion dollars[13]. As It can see inly private companies are building pilot and industrial plant for this purpose. BIONANOPOLYS partners are offering SMEs the possibility to perform and test bioprocesses in their products, promoting valorisation or upcycling of the wates or by-products. Any type of biomass can be studied and treated to get the best value from it. ***Cellulose nanofibers and nanocrystals, nanolignin.***

Cellulose nanofibers Lignocellulosic feedstock such as agricultural waste, wood wastes and forestry residues are mainly composed of cellulose, which is considered intermediate products constituting the base to produce CNF and CNC. The recalcitrance of lignocellulosic biomass due to the interaction between cellulose and hemicellulose along with the degree of delignification requires a pretreatment process to separate the main component of lignocellulosic material. Based on the literature, several steps have to be followed for the valorization of this component in order to obtain CNF, and these include fibers fractionation by a mechanical and/or chemical treatment. The most popular method is based on the mechanical fibrillation process, in which the cellulose is applied to a high shear to break the aggregates, giving as result fibrils around 20nm in thickness and few micrometres in length. From an economic point of view, this is the most suitable method[14]. Currently, NFC production by **homogenization is available at pilot scale at CTP** (TRL7; 30 kg/day capacity) and leads to a product with a constant and high-quality level. A protocol patented by CTP/FCBA (n° PCT/FR2017/051121) enables to develop an energy efficient homogenization process. Therefore, CTP's ambition in BIONANOPOLYS is to upgrade its **twin-screw extrusion continuous production** process to provide 20% consistency CNF, with flexible production

9 Wyman, C.E., Decker, S.R., Himmel, M.E., Brady, J.W., Skopec, C.E. & Viikari, L. Capítulo 43: Hydrolysis of Cellulose and Hemicellulose. En: Dumitriu, S. Polysaccharides: Structural Diversity and Functional Versatility. 2ª Ed. pp. 1023-1062. Marcel Dekker. Nueva York, 2004.

10 Rav indran, R. & Kumar A. A comprehensive review on pre-treatment strategy for lignocellulosic food industry waste. Bioresource Technology, vol. 199, pp.

92-102. Elsevier, 2016.

11 Bennet, J. W. Mycotechnology: the role of fungi in biotechnology. Journal of Biotechnology,vol. 66, pp 101-107.2018

12https://www.basf.com/th/en/who-we-are/organization/locations/europe/german-companies/BASF_Venture-Capital/portfolio/Renmatix.html.

13 McKinsey &Company releases New Data about Growth in Industrial Biotech Sector at World congress. 2015.

14 S. Laurichesse and L. Avérous, "Chemical modification of lignins: Towards biobased polymers," Progress in Polymer Science. 2014.

capacities from 2kg/h to 20kg/h. On the other hand, **disintegration process** is available at pilot scale at ITENE facilities. During BIONANOPOLYS the implementation of a continuous process will be carried out at ITENE pilot plant increasing production capacity from 1kg/day to 2 kg/day dry CNF. In BIONANOPOLYS project, is proposed to improve the productivity of the pilot plants optimizing the process of both mechanical fractionation of biomass to obtain cellulose nanofibers. Both CTP and ITENE make available to this project the knowledge acquired in previous projects on biomass processing and cellulose nanofibers to improve and optimize the current process by adapting their pilot lines with a set up for a continuous production process. Cellulose nanofibers have great potential to be chemically modified and reengineered into a potential application that can range from papermaking, coatings and films, nanocomposites, pharmaceuticals, separation membranes, rheology modifiers, oil and gas, or even biomedical engineering[15].

Cellulose nanocrystals: The process of extraction CNC will be performed by an acid hydrolysis treatment and after diluting the CNC and centrifuging to remove excess acid, the precipitate will be then dialyzed with water to remove non-reactive sulphate groups, salts and soluble sugars, until reach neutral pH. By treating cellulose with concentrated acids known as acid hydrolysis, the amorphous regions can be broken up, thereby producing nanosized cellulose-based crystals called nanocrystalline cellulose (NCC) or cellulose nanocrystals (CNC). CNC are elongated, rigid and rod-like or whisker shaped particles with a rectangular cross-section. It has a high aspect ratio (length to width ratio) with typical lateral dimensions of 100-200 nanometers (nm) and longitudinal dimension of 5-20 nm. Their dimensions vary depending on the native cellulose source, extraction methods and recovery processes such as hydrolysis time and temperature. CNC are interesting because have high tensile strength and can be compared to Kevlar fiber in stiffness, making it an excellent material for reinforcement of natural or synthetic matrix polymer[16] and have unique rheological properties. The dispersions CNC at low concentrations provide high viscosity making it a great ingredient as a non-caloric stabilizer for food industry. In addition, CNC are highly shear thinning (a property which is particularly important in different coating applications as well as an ideal value-added biomaterial with potential for utilization in stimulation, fracturing and completion fluids)[17]. CNC in composite films has been suggested to act as a barrier material for perishable foods and materials which are sensitive to air and oxidation. These characteristics make CNC a very important material to obtain in this project. In BIONANOPOLYS the main challenge for cellulose nanocrystal production will be the optimization of the production process in terms of yield, chemicals, energy and water consumption and wastes to satisfy the increasing demand of manufacturing new types of cellulose nanocrystals based materials on an industrial scale.

Nanolignin: Lignin nanoparticles (NL) are sustainable, green material which can be derives from industrial and agricultural waste. NL has enhanced properties compared to standard lignin due to a larger surface to volume ratio and are suitable for a wide range of applications including drug delivery, stabilizing agent, and reinforcement materials[8]. Many approaches have been reported to synthesize NL which include chemical methods (solvent shifting, pH shifting, carbon dioxide antisolvent and cross-linking/polymerisation), mechanical methods (homogenization and ultrasonication) or other methods such as ice-segregation, template base synthesis, aerosol processing and electrospinning[9]. Among the above methods, the most investigated are solvent and pH shifting. However, both approaches require a massive amount of solvents and involve downstream processing which may not be cost effective in the industrial scale. For this reason, in BIONANOPOLYS, the nanolignin will be prepared via

[8] Perera, U. P., Foo, M. L., Tan, K. W., & Chew, I. M. L. (2019, October). Process Modelling and Economic Evaluation for NanoLignin Production. In IOP Conference Series: Materials Science and Engineering (Vol. 652, No. 1, p. 012054). IOP Publishing.

[9] Beisl, S., Miltner, A., & Friedl, A. (2017). Lignin from micro-to nanosize: production methods. International journal of molecular sciences, 18(6), 1244. 20 Feldman, D. (2016). Lignin nanocomposites. Journal of Macromolecular Science, Part A,53(6), 382-387.

an ultrasonication treatment at different frequencies and oscillation amplitude and treated for different times to optimize the process and make it more cost efficient. The presence of nanolignin in different polymer nanocomposites developing in this project can contribute to: strength and thermal stability improvement, antibacterial activity, flame retardancy, recovery of Ag ions and adsorption of Pb(II) ions, oil-spill clean-up processes packaging and other properties[20].

15 S. H. Osong, S. Norgren, and P. Engstrand, "Processing of wood-based microfibrillated cellulose and nanofibrillated cellulose, and applications relating to papermaking: a review," Cellulose. 2016, doi: 10.1007/s10570-015-0798-5.

16 . George and S. N. Sabapathi, "Cellulose nanocrystals: Synthesis, functional properties, and applications," Nanotechnol. Sci. Appl., 2015, doi: 10.2147/NSA.S64386.

17 R. J. Moon, A. Martini, J. Nairn, J. Simonsen, and J. Youngblood, "Cellulose nanomaterials review: Structure, properties and nanocomposites," Chemical Society Reviews. 2011, doi: 10.1039/c0cs00108b.

Building Blocks and Block copolymers.

Product derived from fermentation include lactic acid, succinic acid, polyhidroxyalcanoates (PHA), et. Lactic and succinic acid have many applications in the food, chemical, textile and pharmaceutical industries, and recently there has been a growing demand of these compounds in the manufacture of the biodegradable polymers, polylactic acid (PLA) and polybutylene succinate (PBS), respectively.

Block copolymers have traditionally been synthesized from petroleum-based polymers, being applied in different sector as medicine, packaging, electronics, etc. In these copolymers, each block is able to keep its properties, thus differences in crystallinity and miscibility with the matrix to reinforce, produce a separation between microphases. As a result of this separation and the ability of self-assemble of the building blocks, they are ordered into **nanostructures**, that can be modified by changing molecular weight, composition, type of ordering, etc., giving rise to different properties[21]. This versatility enables the obtention of tailor-made products and positions block copolymers as substances of great interest in the field of advanced materials.

However, the use of **bio-based building blocks** to obtain block copolymers is not widespread, except for the medical sector, where further research has been carried out in recent years, based on their stimuli responsive potential[10]. Meanwhile, the interest in the use of bio-based materials for packaging applications has increased, reaching a 53 % of the total market of bioplastics[11]. As a result, many lines of research have come up, with the aim of improving their properties and processability, including the use of block copolymers as a nanoreinforcement.

Block copolymers are usually synthesized by living polymerization, which is a form of chain growth polymerization where chain transfer and termination are absent, what makes the presence of a monomer the only factor to control the growth of the chain. Thus, the chains grow at the same average rate until all the monomer reacts, resulting in polymers with narrow molecular weight and controlled length. This polymerization can be performed following the mechanisms of typical polymerizations: anionic, cationic, ring opening, etc. or even combination of them [12, 13].

The block copolymers selected are synthesized in bulk by **ROP (ring opening polymerization)**, having a prepolymer (PBS and PHBV oligomers) as macroinitiators and in presence of a suitable catalyst[14, 15]. In this case, the reaction is an anionic ROP, initiated by these macroinitiators, followed by an α route of propagation and transferred as long as monomer is available[16]. On the other hand, bulk polymerizations give rise to other problems, as difficulty to keep

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enough degree of mixing when molecular weight and viscosity increase, or lower productivity due to incomplete reactor discharge. The difficulties of bulk synthesis will be overcome by upgrading and adapting a polymerization reactor to keep mixing degree during reactions and developing a fast characterization protocol to reduce trials time (for this key characterization equipment is locates in the same place as the pilot plant (ITENE)).

BIONANOPOLYS will develop different innovative biobased products by taking advantage of the versatility of these block polymers, customizing them for each product, focusing specially in food and cosmetic packaging. Thus, superior barrier, mechanical properties, thermal resistance or processability will be achieved to meet the requirements of the final application and, also, the legislation and market needs.

Metallic nanoparticles

Metallic nanomaterials, including metallic nanoparticles (NP), have received a huge attention due to their unique catalytic, electrical, magnetic and thermal properties[17]. Due to its bactericidal effect, silver NP (AgNP) have been studied in the last years.

Bacteria and fungi represent prokaryotic and eukaryotic microorganisms as the main bioinspired candidates for MNM synthesis. There is a growing knowledge about the resistance of metals in the microorganism-mediated

21 Hadjichristidis N., Pispas S., Floudas G. Block Copolymers: Synthetic Strategies, Physical Properties, Applications. Wiley-Interscience, 21, 385 – 408, 2003.

bioreduction process. Several enzymes are believed to be actively involved in the bioreduction process of transporting electrons from certain electron donors to metal electron acceptors for different microorganisms.

Since noble metal ions are toxic to microorganisms, the resistance of metals in the microbial reduction of these metal ions must be considered. Both Ag(I) and NP Ag ions are well known toxic elements for living microorganisms. However, some studies showed that the toxic effects of Ag(I) could be minimized by reducing the elemental Ag(I) ions to Ag NP within the cells Thus, biosynthesis of AgNPs can be carried out by using Penicillum fungi. Fungal biomass is separated, and cell-free extract is used to AgNPs preparation. Water extracts from selected plants contain reducing and capping reagents. These extracts are used to synthesize nanoparticles[18,19]. Flavonoids (polyphenols), which can be also extracted from leaf of selected plants have shown that are responsible for the reduction of silver ions[20]. To extract these bioactive compounds, ionic liquids are used. This process is very sensitive to different factors like acidity (pH), initial concentration of starting solution or molar ratio.

In BIONANOPOLYS, AgNPs will be synthesized to be introduced in materials for high added value applications. Subsequently, this NP will be used for the functionalization of cellulose nanofibres and nanoclays to optimize their cost/performance ratio. As aforesaid, Ag has shown bactericidal effect at very low concentration, which makes it a potential candidate for applications in the medical sector and food and cosmetic packaging. In this project, efforts will be focused in the improvement of the yield and scale – up of the process to produce enough quantities to carry out pilot trials.

Nanocapsules

Nanoencapsulation regards the technology of enclosing a certain nanomaterial, either solid, liquid or gas within a second material, thus forming capsule on nanometric dimensions with an active core dispersed or trapped inside a matrix or shell[21]. The nanomaterials that constitute the core can be selected from a wide range of specimens (e.g., drugs, perfumes, antimicrobials, antioxidants, saccharides, etc.), with the purpose of meeting the specifications and requirements of a certain application or market segment. On the other hand, the shell material is responsible for the maintenance of the structure of the nanocapsule and integrity of its core. The duration of this protection can be permanent or finite, in which case, its content is often released by diffusion or in response to an external

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trigger (shear stress, pH/temperature change, moisture, enzymatic activity), enabling their controlled and timed delivery to a targeted site [22].

Due to the need to change to a more sustainable and circular economy, the search for alternative raw-material sources and processes has led RTOs and industrial stakeholders to understand the potential of bio-based materials for the development of innovative technological solutions, such as functional formulations, coatings, packaging materials, etc.. In this sense, biomass based materials, such as cellulose and its derivatives are of most interest for the development of new capsule materials for the delivery of selected active agents, due to their sustainability, low environmental impact, inertia, biocompatibility, porosity, among other properties [23], being suitable for the development of controlled release micro/nanocapsules[24]. Within cellulose derivatives, ethyl cellulose has been receiving more attention since the chemical modification of cellulose allows it to interact with other materials and different encapsulated and release systems[25].

The development of functional nanocapsules for the controlled release of active agents is of most interest for several economic sectors, not only for the delivery of the active substances, but also because their submicron dimensions offer many advantages in terms of particles distribution, dispersion and mechanical properties, for example. In BIONANOPOLYS a continuous process will be developed, focused on the nanoemulsion technology

through the homogenization of the emulsions with a higher mechanical stirring and optimizing the downstream processes regarding the centrifugation and drying steps. Multifunctional capsules will be obtained to develop packaging, cosmetics, and agricultural applications, but with potential to be applied in a wide range of applications.

1.4.1.2 Network 2. Bionanocomposites pilot plant.

Modification and functionalization of nanomaterials.

The homogeneous distribution of nanomaterials in polymers is a prerequisite for the superior properties of nanocomposites. However, most of them, are known to agglomerate usually in dispersions or polymer matrices because of strong attraction between them. Surface modification of nanomaterials is a common way to reduce their interactions, enhance the chemical compatibility with polymer matrices and improve dispersion levels on nanomaterials in the polymers. Three approaches can be considered to modify them: liquid-solid and gas-solid reactions and grafting by chromatogeny with in BIONANOPOLYS. Green chemistry will be applied on all of them. Numerous literatures for modification as well as preparation of dispersion with nanomaterials are found based on **liquid-solid reactions**. They normally are carried out in a vessel equipped with mechanical stirring systems where modification takes place, and sometimes followed by ultrasonication for de-agglomerating and dispersing nanomaterials in aqueous based media, necessary to improve homogeneity and stability of the suspension. Several types of nanomaterials, such as cellulose nanofibres[26], sepiolite, carbon nanomaterials, metal oxide nanoparticles[27] and clays[28] can be modified by using these dispersion systems. The high amount of reagents, time of reactions and energy consumption involved in the modification process as well as in the post-treatment process to obtain a high quality additive in powder form, are the main causes involved in the final price of this kind of nanomaterial. Based on the literature, where sonication is often used to break particle agglomerates, increasing the specific surface area, this system will be implemented in BIONANOPOLYS on the top of the reactors to improve the surface functionalization of the nanomaterials, Then, **increasing the functionalization reaction yield, a reduction of reagents, time and energy consumed** could be achieved, with the consequent saving in raw materials, energy and

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[27] Wen-Bin Tsai et al. Journal of Nanoparticles Volume 2016, Article ID 6539581, 9 pages.

[28] Satriananda et al 2018. Journal of Physics: Conference Series, Volume 953, conference 1.

production of chemical residues. With the same objectives, a new mixture technology based on an emulsifier will be set on the bottom of the reactor. This system will allow to work at lower processing temperature, higher solids loading capability and faster, as well as dust-less powder induction.

By other side, to apply also the concept of "green chemistry" in modification processes of nanomaterials to achieve more cost efficient and environmentally friendly nanomaterials, **gas-solid** reactions will be carried out in BIONANOPOLYS project. In this kind of reactions, the use of solvents in the reaction and the post-treatment processes are unnecessary in comparison with liquid-solid reactions. Solid-gas reactions have been involved in surface functionalization of silicon nanowires and other metal oxide nanoparticles[29] mainly used for fields like gas sensors, biosensors, and electronics. In BIONANOPOLYS, this technology will be used to modify nanomaterials in a more sustainable way and evaluate the feasibility of scaling this process for further applications.

Other nanomaterials modification technology is the **Chromatogeny.** The chromatogeny technique create a water repellent surface, is therefore highly interesting for the cellulosic based materials barriers. The surface chemistry of cellulose based materials can be modified, introducing aldehyde and carboxyl groups, which have major potential for surface functionalization. CTP has a patent to do with the modification of cellulose nanofibers. This patented method (patent number WO2012066015A1) consists of treating a hydroxyl substrate by means of a **chromatogenic graft**[30] and CTP makes available to this project all its knowledge to achieve an optimal modification for the final products where cellulose nanofibers will be incorporated or applied. CTP optimized this process to treat cellulosic web materials (papers, boards and tissue rolls). This roll to roll solvent-free surface treatment process is inspired from vapor phase chemical deposition. Chromatogeny, judiciously takes advantage of the liquid-vapor equilibrium to get an ultrafast acylation of the surface reactive OH groups of cellulose or polymers having such OH groups, yielding in the derivation of the extreme surface of fibres/polymers into corresponding fatty esters. Thus, obtained permanent hydrophobic sheath around lignocellulosic fibres confers water, gas and grease barrier properties while keeping recyclability and biodegradability. BIONANOPOLYS project will focus on the upgrade of this technology to be applied on different nano-cellulosic materials, in particular CNF films.

Nanocomposite compounding in thermoplastic.

Melt compounding is the most promising processing method for nanocomposite preparation in industrial scale. This process offers end-use manufacturers many options to cover final product specifications (e.g. selection of polymer grade, choice of reinforcement, reinforcement content, blends with other additives, etc.), and minimizes capital costs due to its compatibility with existing processes. Co-rotating twin screw extruders are high-speed extruders used primarily in compounding applications.

However, melt compounding also presents some challenges, such as, the dispersion of the reinforcement in the matrix and the feeding of the nanoreinforcements. When solids are compounded into a molten polymer, they are commonly introduced as large agglomerates and aggregates, or solid clusters held together by cohesive forces. Surface treatments of the nanomaterials[31], or compatibilizers[32] can be added to improve the dispersion of the nanoreinforcements and to achieve better interaction with the matrix.

Nanocomposite usually are characterized by offline techniques to get the full picture of the multi-scale morphologies of nanocomposites and dispersion quality of nanoparticles in solid matrices (XRD, SAXS, WAXS, SEM, TEM, etc.). Different measuring techniques have been applied to evaluate the dispersion along the extruder. The use of NIR spectroscopy in the barrel of a twin screw extruder was tested to predict the desired dispersion

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dependent properties of PP and montmorillonites nanocomposites, after initially correlating data from inline method with offline data through multivariate data analysis by chemometrics analysis[33].

Besides this, specific techniques such as on-line rheometry and in-line near-infrared spectroscopy (NIR) [34], dielectric[35] optical[36] or "*Terahertz spectroscopy*" (measure of additive content and melt density)[49], some of them have been applied to monitor the evolution of dispersion along a twin screw extruder. To assure the optimal dipersion of our nanoparticles in the polymer matrices two in-line techniques will be set up in the compounding process, rheology and optical techniques, that will allow us to get the optimal process parameters faster than with the previous offline characterization. Trials time is foreseen to be reduced at least 50%, and accuracy of the process parameters will be also improved by 20 %.

Biobased nanodispersions

In order to prepare **stable dispersions of nanomaterials** for different applications, typical equipment used to dissolve polymers into solvent are based on low speed propeller, turbine or rake type agitator. At ITENE, high shear mixers have been used to prepare polymer system in liquid state, which generates significant savings in power consumption because of the reduction in mixing time (around 80%)[37].

This technology will be used for preparing dispersions from nanomaterials in different solvents as well as polymers matrices for coating applications, improving the dispersion level and stability of different types of nanomaterials in a wide range of solvents and polymers. Moreover, the use of the emulsifier will imply reduction of mixing times, increasing the production rate of nanodispersions, from 20 l/hour to 50 l/hour as minimum, as well as reduction of energy consumption by working with lower processing temperatures and dust-less powder induction. For those cases where dispersions need to be carried out in polymer solutions, the ITENE's pilot plant will be equipped with a new reactor where high shear mixer will be set on the top of the reactor, reducing time and solvents volume from the cleaning step.

The goal of **in situ polymerization** is the insertion of the nanoparticle in the polymer matrix during the polymer synthesis process. This method can be made in a single step when polymeric chains are growing thus increasing the compatibility and the dispersion of the nanoparticle within polymer matrix. This approach has proved better dispersion that melt compounding and consequently improved properties (mechanical and barrier), even with

lower amounts of nanoparticles, has been observed. Considering these results, during the last years CIDAUT has studied the formation of nanocomposites based on different polymer matrices (polyolefins, polyamides and polyesters) catalytic systems and nanoparticles. Nowadays all the **industrial polymerization lines** producing raw materials (REPSOL; SABIC; etc.), and the manufacturers of polymerization catalysts need smaller pilot lines to test new technologies (catalysts, nanoparticles) and design custom made materials for short series production, before upscaling the technology to larger reactors working in continuous production. There are more and more industries demanding these services and the research carried out by CIDAUT in **in-situ polymerization techniques** has enabled a know-how, that allowed the creation of a polymerization pilot line to fill a niche between small lab scale tests and huge polymerization industrial plants. During BIONANOPOLYS project, CIDAUT will adapt this technology to have a more versatile *in situ* polymerization line for obtaining bionanocomposites as alternative to their petroleum-based counterparts[38]. The upgrade of this pilot line will consist on changes in feeding systems, for solids and gases, and stirring systems to allow strict control of the polymerization environment. These improvements will enable an

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[38] L. Martino et al. Influence of organically modified montmorillonite and sepiolite clays on the physical properties of bio-based poly(ethylene 2,5furandicarboxylate). Composites B, 2017, 110, 96-105

increase in the production rate as well as offering a wider service that will include different products and knowledge in obtaining *in situ* bionanocomposites.

1.4.1.3 Network 3. Bionanoproducts pilot plant.

The purpose of network 3 is to produce products reinforced with the nanomaterials above described and validated for different applications. Each PP partner owns high expertise and background in these techniques, so the companies will benefit of this knowledge when developing new products.

Rigid and flexible packaging.

Different conversion processes will be used to validate materials developed for packaging applications:
Cast extrusion (monolayer / multilayer) polymer(s) are melted from multiple feed streams and introduced into a feed block and then into a single-slot die or they can be introduced into multi-manifold and then multi-slot die, resulting in both cases in a structure of overlapped layers. In multilayers, additives or nano-additives are introduced in the intermediate to control release or avoid direct contact with the food/cosmetic product. When the nature of the materials is very different, an additional tie layer or adhesive is applied at high temperature to bond them. Thermoforming: when thicker layers are processed by the aforesaid technique, they can be converted in rigid packaging by thermoforming processes. This technique consists of heating plastic sheet to a softened, pliable state, then pressing the sheet against a cool mould, taking the desired shape.

Cellulosic products.

Two types of cellulosic will be produced in BIONANOPOLYS project: paper and CNF-polymer standalone films.

Paper production or papermaking is a two-step process in which a fibrous raw material is first converted into pulp, and then converted into paper, by mechanical or chemical processes. This pulp is bleached and further processed, depending on the type/grade of final paper. In the paper factory, the pulp is additivated, to obtain different formulations depending on the application. Formulations are dried and pressed to produce paper sheets. The additives can be added into the pulp suspension in bulk at the beginning of the process or as a coating onto the final paper. The addition of BIONANOPOLYS nanoadditives will improve the properties and confer new functionalities to the final materials.

CNF-biopolymer films will be produced using a process analogous to paper making. Instead of cellulosic pulp, a CNFbiopolymer and biobased nanofilaments will be used and transferred on a hydrophobic filter. The obtained standalone films will be used in 3 different ways: 1) as such, 2) as barrier layer on/into multimaterial products or 3) shred and incorporated into polymer as charge, after hydrophobisation of the film (see chromatogeny in 1.4.1.2).

Textiles and non-woven fabric.

BIONANOPOLYS will work on different technologies to produce textiles and non-woven fabric for applications like personal care or cleaning products. Production capacity and flexibility will meet customer's most demanding needs. Monofilament extrusion is broadly used in industry, to make strands of polymers for different applications. Coextrusion/bicomponent extrusion combine various polymers to confer different functionalities to the final product. Nanobubbles technology is used for textile applications to finish garments. It uses air to produce a micronized state of chemical compounds, then are transported via nanobubbles and injected into a washing machine where the

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garments are located, and the chemicals entrapped, impregnate it. This technique is a faster and more sustainable way for garment finishing/dyeing than traditional systems (e.g. dip coating) in terms of water/energy savings (80 – 90%) and can process several chemicals (water repellents, antimicrobials, softeners…) and dyes, with processing conditions close to industrial scale.
Melt blown process is a non-woven manufacturing system involving direct conversion of a polymer into continuous filaments, integrated with the conversion of the filaments into a random laid non-woven fabric. It will be used to obtain improved filament fineness (smaller diameter) and electrostatic modified filament surface for filter sector. Other auxiliary technologies are considered in BIONANOPOLYS to improve the efficiency of non-woven production. Carding is a mechanical process that disentangles, cleans and intermixes fibres to produce a continuous web or

sliver suitable for subsequent processing. Needling process forms textiles or fabrics directly joining fibres and bypassing the need for yarn. Meanwhile, stitch bonding consolidates fibre webs with knitting elements with or without yarn to interlock the fibres.

The communication of technical and quality parameters will be feasible at a higher technical level corresponding the fundamental characteristics of an Industry 4.0 environment. By this way, consumption of material will be reduced, and better parameterisation and reproducibility of non-woven products will be achieved.

Injection and 3D Printing

Injection moulding starts with a melting process of polymeric materials produced by friction between a rotating screw and a barrel, the screw advance, injecting the molten plastic into a mould cavity under high pressure. Then, it cools and solidify taking shape of the mould, which is opened automatically and the part is ejected by ejector pins. The injection equipment of ITENE will be optimized to produce customized products with bio-based materials.

3D printing, also known as additive manufacture (AM), is a process for constructing 3D physical objects from digital models through the successive layer-by-layer deposition of different materials[52]. There many different 3D printing technologies divided according to the ISO/ASTM 52921:2013 such as material extrusion, material jetting, etc[53]. During this project, 3D printing pilot lines will be updated to allow the process of biobased materials. First, a lab scale filament maker will increase productivity (15%) and capacity of processing high viscosity materials by new screw design. On the other side, two home-made printers will be modified to strict control temperature and moisture, what is important to minimize current problems of high crystallinity and water absorbing, in polar polymers and (nano)composites.

Foaming.

Extrusion foaming is a highly cost-effective and continuous production process to obtain products with different shapes for thermal insulation, packaging, etc. The equipment is a single-screw extruder with a gas injection unit, in which a blowing agent (CO_2, N_2, etc.) is dissolved at high pressure into the molten polymer, then is cooled down quickly and foamed at the exit of a die with the required shape.

Bead foaming is a batch process in which polymer micro-pellets are fed into a stirred vessel together with water and additives, a blowing agent (CO_2) is injected at high pressure and mixed, then pressure is released, and the pellets expand into beads, which are thermoformed with steam into complex shape products.

This technology is used mainly with fossil based materials, but currently there is a high interest in the industry to find sustainable alternatives based on biopolymers, such as polylactic acid (PLA) or polyhydroxialcanoates (PHA). However, several drawbacks, as slow crystallization kinetics or low mechanical properties, needs to be overcome, for which the addition of different biofillers is studied[54].

Inorganic nanoparticles, like nanoclays, play a **multifunctional role**, since the nucleation of cells during foaming is strongly influenced by the presence of nanoparticles (**heterogeneous cell nucleation mechanisms**) and their intrinsic reinforcing effect and capacity of induce morphological changes result in an **improvement of mechanical properties**[55]. Focusing on BIONANOPOLYS food packaging applications, should be noted that nanoclays usually enhance barrier and antimicrobial properties, what contributes to extend the shelf-life of the packed product[56].

52 Bhatia, S. K., & Ramadurai, K. W. (2017). 3D Printing and Bio-Based Materials in Global Health. Switzerland: Springer International Publishing AG.

53 Liu, J., Sun, L., Xu, W., Wang, Q., Yu, S., & Sun, J. (2019). Current advances and future perspectives of 3D printing natural-derived biopolymers. Carbohydrate polymers, 207, 297-316.

54 Nofar, M. Effects of nano-/micro-sized additives and the corresponding induced crystallinity on the extrusion foaming behavior of PLA using supercritical CO2. Materials & Design, 101, 2016, 24–34.

55 Lai, S.M.; Huang, C.Y.; Li, S.C.; Chen, Y.H.; Hsu, H.C.; Yu, Y.F. and Hsiou, Y.F. Polymer Engineering and Science. 51, 434-444.2010.

56 Darie, R. et al. Effect of Nanoclay Hydrophilicity on the Poly(lactic acid)/Clay Nanocomposites Properties. Industrial & Engineering Chemistry Research, 53(19), 2014, 7877–7890.

When fibres are introduced, play a key role in the formation of a uniform cellular morphology, limiting the expansion ratio of the composites. This results in improvement of mechanical properties and reduction of weight[57].

Coatings applied onto surfaces

Nowadays, there is a trend to use "greener" packaging materials as the European Union strives form more sustainable packaging. To get similar properties to conventional materials, biobased ones are usually coated to enhance surface properties, print quality, improve barrier or to incorporate new active functionalities.

Spray coating: Thermal spray coatings are formed by successive impact of a stream of spray droplets in fully or partially molten states, followed by flattening, rapid cooling and solidification. Nozzles of the spraying system are specifically selected for each system (coating – substrate). 2D & 3D spraying equipment will be adapted to the customers necessities.

Roll to roll (flexo and gravure): flexo printing, is a relief printing technique, whereby the ink is transferred from a raised printing plate using fast drying inks onto the chosen substrate. Gravure printing or rotogravure is an intaglio printing process that uses rotary printing capable of producing continuous tone images. Large copper plated steel cylinders are engraved with the chosen design using a laser or diamond tipped stylus.

Coating thickness and homogeneity is crucial to reach superior barrier properties. In-line monitoring system will be set up to monitor application of materials and quality, reducing trials time from days to minutes and increasing product quality, optimizing industrial parameters.

Electrospinning: also known as electrodynamic spraying, can produce tiny droplets with submicron sizes by means of an electric field. Electrospraying can also be used to produce fine polymeric particles, which are widely used for biomedical applications, particularly drug encapsulation. Production process is still not well introduced at industrial scale and, therefore, our pilot line will allow the companies to apply this technology and will test their results.

1.4.2 State-of-the-art and beyond: BIONANOPOLYS user cases.

As a result of the upgrading and different bionanomaterials developed in BIONANOPOLYS pilot plants, eight industrial partners will be able to bring to market 21 improved products. These end-user cases and the degree of innovation expected are gathered in the following table.

Table 2. BioNanoPolys User cases used to validate the technologies of the different pilot plants.

Bionanomaterial used / Product	Objectives, targets	Pilot plants or services
textisol non woven CNF, active nanocapsules, metallic nanoparticles	- UC1: Cellulose nanofibers will be used as reinforcement of non-woven textiles for personal care and cleaning wipes to improve the final product to make them more environmentally friendly. Objective: 100% compostable product - UC2: Active nanocapsules with antimicrobial activity provided by the essential oils nanoencapsulated to reduce odours and antimicrobial effect in textiles for personal care (Non-woven textiles).	PP1, PP5, PP10, PP11 Services: compostability, mathematical modelling
DS Smith CNF, active nanocapsules.	- UC3: Cellulose nanofibers will be used as reinforcement of paper-based materials in terms of mechanical and barrier properties. Improvement up to 40% and 60%, respectively, are expected to achieve. - UC4: Chemically modified and hydrophobic cellulose nanofibers will be used with the aim of obtain new functionalities to the paper materials. Water and grease barrier are expected to reach. - UC5: Paper with active properties (antimicrobial and antioxidant) will be developed with the applications of nanocapsules to paper materials by coating. Improvement of barrier properties up to 30% and reduction of antimicrobial load in food products, and oxidation delayed at least 20%.	PP1, PP2, PP6, PP8, PP10, PP14. Services: paper recyclability, in – line monitoring
NOVAMONT PLA-PBS and PHBV-PLA block copolymers, CNF, Nanocapsules, nanoclays/	-UC6: Block copolymers will be used as reinforcement of biobased / compostable formulations for film packaging, injection moulded cutlery and coffee capsules, and thermoformed trays and containers. It is expected to improve up to 20% thermal resistance, 30% mechanical properties and 20% barrier properties of polymer matrices. - UC7: Chemically modified and hydrophobic cellulose nanofibers will tune the rheology (i.e. viscosity and filmability) of water-based dispersion to be applied	PP1, PP2, PP3, PP5, PP6, PP7 Services: mathematical modelling, In-line monitoring.

57 Ventura, H et al., Gas Dissolution Foaming as a Novel Approach for the Production of Lightweight Biocomposites of PHB/Natural Fibre Fabrics. Polymers, 10, 3, 2018, 249.

Bionanomaterial used / Product	Objectives, targets	Pilot plants or services
Biodegradable and/or Compostable Packaging	as barrier coating on paper-based substrates, improving barrier properties up to 50% and mechanical properties up to 30%. -UC8: Nanocapsules will confer various functionalities, e.g. antimicrobial features resulting in extended up to 30% shelf-life of packaged food and cosmetics. - UC9: Nanoclays will reinforce the biodegradable formulations, increasing up to 40% mechanical and 25% barrier properties of 100 % compostable non-food packaging carrier bags to be reused for organic waste collection.	
CNF /CNC, active nanocapsules/Biobased personal care formulations	- UC10: CNF/CNC will act as film forming agent in cosmetics formulations, substituting fossil-based additives. - UC11: Nanocapsules will act as delivery systems being a vehicle of active agents to provide, i.e. antimicrobial, antioxidant, antiwrinkle and moisturizing properties to cosmetics formulations with measurable improvements with the aim of replacing chemically produced delivery systems. Objective; same behaviour than actual cosmetics formulations: shelf life of at least 2 years. - UC12: Functional food will be developed by the application of active nanocapsules to different food products.	PP1, PP2, PP5, PP6, PP8. Services evaluation of compliance to EU regulation 1223/2009 for cosmetic application
Logoplaste Block copolymers (PLA-PBS and PHBV-PLA), CNF, modified natural clays and active nanocapsules or metallic nanoparticles and their nanocomposites / Compostable packaging for food and personal care applications	LOGOP will develop two products of rigid packaging. -UC13: bottle for personal care products injection stretch blow moulding (ISBM) reinforced with active nanocapsules with the aim of increasing the shelf life of the packaging solution. Shelf-life extension up to 30%, respect to biodegradable base material. -UC14: bottle for food applications (e.g. sauces) performed by co-extrusion blow moulding with biodegradable biopolymers reinforced with nanocellulose and/or nanoclays to improve barrier. Reduction of OTR and WVTR up to 3040% respect to biodegradable base material. Both packages 100 % compostable	PP1, PP2, PP3, PP5, PP6, PP7, PP12. Services: food safety and compostability tests.
CellMat Technologies CNF, nanolignin and nanoclays / Foamed products	Matrices of PLA and PHA with different nanoreinforcements to produce: -UC15: Thermoformed foamed trays for food packaging, with improvements of up to 20% extended shelf life and 20% mechanical properties (or 20% weight reduction at the same mechanical performance). -UC16: Foamed packaging based on bioexpanded beads, improvements of 10% in thermal resistance and 20% mechanical performance (or 20% weight reduction at the same mechanical performance) are expected. -UC17: Complex shaped foamed parts for the automotive sector with improvements similar to previous product.	PP1, PP2, PP5, PP7, PP14 Services: rapid prototyping

danipack Block copolymers (PLA-PBS and PHBV-PLA), CNF, active nanocapsules, modified nanoclays / Compostable and a recyclable packaging for food applications	DANIP will develop two products of flexible packaging for food applications, 2 laminated film ranging from 6 to 200 µm to be thermoformed for vacuum packed applications (meat and cheese), with two different objectives: -UC 18: a recyclable solution based on non-biodegradable biopolymers, e.g bioPA, bioPE, bioPET reinforced with nanocellulose and/or nanoclays to achieve high barrier performance, making possible to reduce the number of layers in the system. Also, active nanocapsules will be incorporated to release active agents with antimicrobial and/or antioxidant properties to increase the shelf life of the food. Reduction of OTR and WVTR up to 20-30% and active packaging to extend shelf-life up to 20%, respect to biodegradable base material -UC19: a compostable solution based on bionanocomposites developed by using biodegradable polymers reinforced with nanocellulose and/or nanoclays to achieve high barrier performance and with active nanocapsules with the aim of increasing the shelf life of the packaging solution. Reduction of OTR and WVTR up to 15-25% and active packaging to extend shelf-life up to 20% respect to non-biodegradable material, to reduce the number of layers or thickness in multilayer systems.	PP1, PP2, PP3, PP5, PP6, PP7, PP9, Services: In-line monitoring, food safety, compostability tests and recyclability studies.
	HTSM will work in the development of two user cases:	PP2, PP5, PP7, PP8, PP11
Bionanomaterial used / Product	Objectives, target	Pilot plants or services
hilos técnicos s.m. High Knowledge in Synthetic Fibers CNF, active nanocapsules, compostable synthetic fibers	-UCM20: synthetic fiber produced with biodegradable materials reinforced with CNF by reactive extrusion for agricultural applications. 100% compostable and biodegradable in soil at least 80%. -UCM21: synthetic fibers produced with biodegradable materials and coated with active nanocapsules including agrochemical to be released in a controlled manner. 100% compostable and biodegradable in soil at least 80%, reducing conventional agrochemical treatments by 20%.	Services: Mathematical modelling, compostability

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2. Impact

2.1 Expected impacts

Open and upgraded facilities at the EU level for the design, development, testing, safety assessment, and upscaling of nano-enabled bio-based materials, easily accessible to users across different regions of Europe.

We are putting together within BIONANOPOLYS existing and consolidated pilot plants developed in several projects (see Table 1 and section 4). The participation of end-users will help to identify regulatory requirements and regulatory barriers to market introduction, and therefore to customize the products design towards the overcoming of existing market barriers. Environmental assessment, nanosafety, and nanoquality, especially for food contact applications will be overcome. Market deployment of new products, business opportunities, funding schemes, result exploitation and risk and feasibility assessment will be as well carried out. BIONANOPOLYS will be present in 12 countries, and accessible easily through the SEP in the main official languages from participating countries in short terms and extending to most of official languages in Europe.

KPI: 14 upgraded pilot lines and a Service Portfolio including environmental, safety, regulation, legal, business, access to public and private finance.

At least a 20% increase in the number of new SME users for existing test beds.

A key aspect to build the consortium has been the number of SME clients in the last five years by each partner in the target sectors. Pilot plants owners have attended more than 450 SMES in the sector from the pilot plants. The transversal technological providers have attended 60 SMEs ion the field of monitoring, characterization, and modelling. Technological transversal service partners have more than 100 SMES clients for LCA, regulation and safety analyses. A communication and exploitation plan has already defined to ensure an optimal stakeholders' engagement during the execution of the project, that will lead to potential clients (section 2.2). National brokerage events are envisaged within

the Task 2.4 Stakeholder engagement. Moreover, 2 key partners will attract SMEs on the one hand EBN is a network of around 150 quality-certified business and innovation centres and 70 other organisations that support the development and growth of innovative entrepreneurs, start-ups, and SMEs. EBAN is the pan-European representative for the early stage investors (over 150 member organizations from more than 50 countries, representing a sector estimated to invest 7.5 billion Euros a year). Indeed, EBAN has given support to 1,500 accelerated and mentored companies under the scope of EBAN events, congresses and activities hosted within or independently from European projects.

KPI: increase this network at least up to 30%: add to our current clients, new 150 SMEs users of the pilot plants and 55 SME users of the service portfolio.

At least 15% improved industrial process parameters and 20% faster verification of materials performance for highly promising applications.

Pilot plants will improve the production processes and will optimize the verification of the quality of the products obtained. A fast characterization method will be also installed, to optimize the products and contribute to achieve this goal. Besides, each pilot line includes each own techniques of characterization which will reduce the time of verification of new products. IRIS, CEA and IMT will offer their technologies for reducing the verification on the dispersion quality and the homogeneity of the coatings from days to less than minutes provided that a precalibration is available for similar materials compositions. Increase of production efficiency, machine parameters more precise and resulting effects recorded faster for nonwovens products will be achieved upgrading the pilot plant in terms of control unit and digital interfaces (control system with novel drive regulators, SPS control and data storage, drive systems with motors/gear motors for the carding system and servo drive for the crosslapper, an online monitoring of the product quality, safety systems to protect workers and machinery, user interface with modern data visualisation, management of recipes and remote data transfer and predictive maintenance). Finally, the user cases will prove the improvement of the process parameters and its performance. Finally, it is important to remark again that it is envisaged to work together with other test beds working on characterization (such as NMBP-07) in order to stablish alliances, test their monitoring/characterization technologies in our products, and offer broader solutions to our future clients recommended these OITBS.

KPI: reduction of at least a 50% the time of performance verification of dispersion grade of nanomatAssociated with document Ref. Ares(2020)6389918 - 05/11/2020erials in

bionanocomposites as well as the thickness and homogeneity of thin layers by including in line monitoring systems in compounding and coating pilot plants, respectively.

KPI: reduction of at least a 20% the time of verification of cellulose nanofibers and compounds quality by online monitoring of the quality parameters.

KPI: 40% reduction of trials number for bionanocomposite production with required properties by modelling and simulation as well as rapid prototyping through 3D printing of nanocomposite-based structures.

KPI: 40% reduction in characterization time as each pilot line includes their own techniques of characterization which will reduce the time of verification of new products in combination with benchmarking of technologies to propose the most appropriate ones, with scientific and technical support.

At least 20% improvement in industrial productivity, reliability, environmental performance, durability, and reduction of life-cycle costs of these materials

BIONANOPOLYS presents an approach totally aligned with sustainability, where both the use of biomass from waste from different sectors, thus avoiding a loss of resources, and landfill disposal are avoided. The proposed processes, such as enzymatic treatment and bioconversion by fermentation, are sustainable processes, which take place under mild process conditions and which are increasingly being proposed as sustainable and efficient alternatives to produce products of interest. In this way, in addition, replacing the raw materials traditionally used by large producers of lactic acid, succinic acid or PHA, which use sugar cane or beet, by other raw materials with great potential, avoiding the shortage of that biomass and widening the use of other biomass sources. BIONANOPOLYS will apply the Eco-design approach to design products with special consideration for the environmental and human impacts, energy efficiency and performance of the product during its whole life cycle to obtain the best solution for the environment and most efficient. Recyclability and biodegradability (industrial, home, sea) will be considered. Further we will consider the LCA, LCC and SLCA. The automatization of production processes and scaling up the capacity of reactors will increase the production capacity of nanomaterials at least 25%.

The user cases will prove the improvement of the materials parameters and its performance in terms of shelf-life extension, or additional functionalities.

KPI: Conversion of biomass into building blocks and PHAs with an efficiency > 75%.

KPI: reduction of at least 30% reagents, water, and energy consumption by incorporating high efficiency stirring and feeding systems (sonication probe, emulsifier, high shear mixers) to increase the modification reaction yields and the solid content per batch of reaction

KPI: Functional additives will ensure the increase of durability of the products. Logoplaste will produce cosmetic packaging with the same shelf-life of actual packaging (from 1 year to at least 2 year), accordingly Ambrosialab will develop a cosmetic formulation with biobased active compounds, Danipack will enlarge the shelflife of their food products by at least 20%. *KPI:* Improvement of mechanical properties will lead in a reduction of polymer weight per unit, what is translated in an improvement of industrial productivity. Cellmat will develop a foamed nanocomposite with 20 % improvement in mechanical properties what means 20% in weight reduction, and also 20% of increased industrial productivity. Also, DS Smith will improve mechanical properties by 40 %, increasing productivity and reducing raw materials and energy consumption.

KPI: Reduction of 20% of GHG emissions associated to the biomass pre-treatment and bioconversion processes.

KPI: 100 % compostable and recyclable products will be developed. Reinforced paper-based user case from DS Smith will validate that technologies applied are compatible with fiber to fiber recyclability. Biodegradable wipes (Textisol), packaging (Logoplaste, Danipack), and synthetic fibers (HTSM) will validate that the materials are 100 % compostable.

Specific objectives and KPI are shown in the section 1.4.2 (State-of-the-art and beyond: BIONANOPOLYS user cases).

Facilitating access to finance (for SMEs in particular) for investing in these materials or in applications using them;

A dedicated WP has defined to ensure access to finance (WP9). The users of BIONANOPOLYS will have access to several instruments within the EASME and EUREKA, ERDF and EFIN to access to new markets, always according to the MRL and TRL and the company. Leaded by EBN and EBAN BIONANOPOLYS will include a service to accelerate the time to market of projects and companies (SME, start-ups and large industries, etc.) such as venture capitals and Business Angels recommendations. EBAN will be holding training methods with investors and Pitching Events to facilitate access to finance and investor communities. EBN will hold Business mentor programmes. G&S will oversee legal aspects. External instruments will be followed for raising funds such as https://dealflow.eu/.

KPI: 20-30 founders and innovative SMEs participating in each workshop, making up over 150 supported projects during the period of action. 10-20 companies participating in each Pitch Event, making up about 100 companies introduced to the investor community over the duration of the action.

Identification of regulatory requirements and regulatory barriers to market introduction.

BIONANOPOLYS will guide industry towards a safe and efficient use of bionanotechnology, especially within SMEs capabilities. The members of the project will incorporate the knowledge and experience of the EU nanosafety community and task force experts from relevant projects (i.e ProSafe, NANoREG, Nanocommons) and international bodies (OECD Working Party of Manufactured NMs), defining proven testing methods, datasets, models, guidance documents, decision trees, and other methods useful to regulators and other stakeholders. Contacts will be established with ECHA NanoMaterials Expert Group (ECHA-NMEG), EFSA and other appropriate institutions to keep up to date. In addition, Safeby-Design (SbD) concept will be implemented, promoting well-balanced strategies that address the product/material design and functionality together with the safety aspects for humans and the environment. BIONANOPOLYS will stablish a data sharing and information exchange platform based on the information exchange process applied under the REACH framework to support risk assessment in the FCM authorisation, which will increase chance of success in the authorisation process.

KPI: a library of 20 well-balanced SbD approaches to design out hazard and exposure and validated in case studies, a new database including high-quality data on the (eco)toxicological effects and human exposure under realistic uses at all stages of the life cycle, and refined testing services and tools for a safety driven decision.

2.1.1 Other expected impacts

2.1.1.3 Social impact

BIONANOPOLYS will have a strong impact on society by delivering sustainable, safe and added value products. The project will respect the provisions of the European Parliament's resolution of 24 April 2009 on regulatory aspects of nanomaterials, according to which the benefits of nanotechnologies should be achieved in a safe and responsible manner and with full consideration of their potential EHS implications. The project is also aligned with the EU 2020 strategy for the creation of growth and jobs. Specifically, the project will contribute to more EU investment into nanotechnology RTD, which will create more jobs by encouraging innovations in several sectors (i.e. functional and structural materials, active ingredients, food, construction, energy, and health).

2.1.1.2 Economic impact

The EU has declared the bio-based products sector to be a priority area with high potential. Bio-based products and biofuels represent approximately **EUR 57 billion in annual revenue and involve 300,000 jobs**[58]. The Bio-Based Materials Market by Applications for the period 2017-2026 is valued at EUR 10,000 million in 2017 and has an expected revenue of EUR 85,150 million by end of 2026, growing at a CAGR of 27.1% between 2018 and 2026[59]. According to this forecast, BIONANOPOLYS must target as an objective the 5% of the market share in R&D investments for nano-enabled bio-based materials to ensure an annual turnover of 48M at the end of the period. Thus, to reach out this numbers a solid Business Model an exploitation strategy has been already pre-defined. Next, each of the markets targeted by IOTB BIONAOPOLYS are detailed.

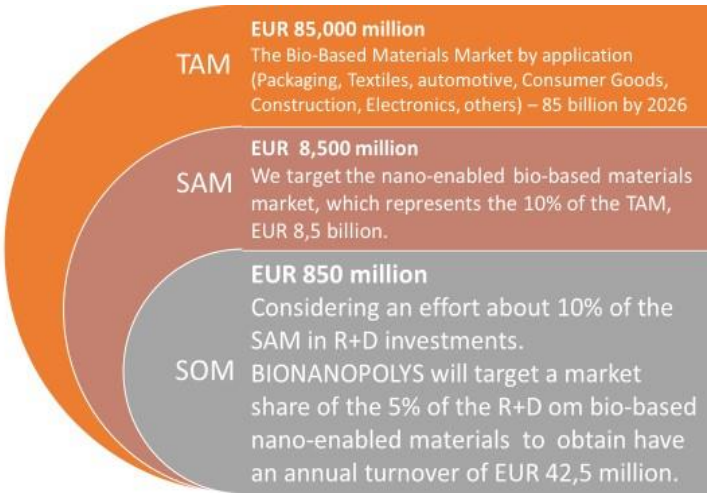

Figure 4. Bio-based nano-enabled market

Table 3: Market size of BIONANOPOLYS sectors.

Markets	Market size and value
Nano-additives, Nanomaterials: Nanocellulose (CNF, CNC), nanolignin, block copolymer,	Nanocellulose and nanolignin: A CAGR of 30.7% is expected for the global nanocellulose market from 2014 to 2025, of which 27.3% will be in paper processing applications, followed by applications for composites and for the food and beverage industry[60],[61]. Biodegradable and/or compostable block copolymers consumption is expected to increase proportionally to biodegradable/compostable packaging market, which is expected to increase by 15% between 2019

[58] https://ec.europa.eu/growth/sectors/biotechnology/bio-based-products_en

[59] https://www.zionmarketresearch.com/news/bio-based-materials-market
[60] Nanocellulose Market Size - Industry Share Analysis Report 2017-2024 (Global market insights).
[61] World Nanocellulose Market - Opportunities and Forecasts, 2017-2023

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Markets	Market size and value
metallic nanoparticles, nanocapsules	and 2024, reaching 2.43 million MT[39]. Main sectors were these nanoreinforcements will be applied are consumer goods, flexible and rigid packaging which are a 40% of the market of these materials[40]. Metallic nanoparticles: Metal nanoparticles market will grow from USD 12.35 Billion in 2017 to USD 25.26 Billion by 2022, at an estimated CAGR of 15.4%. Its application areas and demand for gold nanoparticles in the pharmaceutical & healthcare industry has projected to hold the largest share and dominate the market from 2017 to 2022. This is supported by government initiatives and funding. Its use in electrical & electronics industry is projected to grow at the second-highest rate during the forecast period[41]. Nanocapsules: Micro and nanoencapsulation techniques are increasing, specifically to improve the functionality of the final product. In 2018, the encapsulation market was valued at USD 6.3 billion and a scope of USD 11.08 billion is expected by 2023, with a compound annual growth rate of 11.4%[42].
Cosmetics, active ingredients	The global market: 14.88 billion dollars in 2025, approximately 13.7 billion euros. The added value is in the development of antimicrobial packaging to reduce preservatives in cosmetic formulations.
Functional food: Nanocapsules, active agents	Global market for functional foods, amounted to $ 174.75 billion in 2019. Estimated growth rate of 7.9% from 2016 to 2025, reaching $ 275.77 billion. Its main added value is to reduce preservatives and waste, compostable and safe. Dosage of the active ingredients for better assimilation.
BioNano Composites: Flexible and rigid packaging	While the use of biobased CPBs in medical applications, such as nanoparticles and nanogels for drug delivery, is well established, the development of nanocomposites for biodegradable/compostable packaging enhancement is booming due to increased consumption of these materials. The CPB market is expected to growth proportional to that of bioplastics, whose production is estimated to increase 15% between 2019 and 2024, reaching 2.43 million MT[43]. The sectors in which biobased CPBs, flexible, rigid packaging and consumer goods can potentially be applied, occupy almost 40% of the total market share.
Food Packaging: Flexible and Rigid	By 2026, 31% of the packaging market for food contact is expected, reaching 7.5 Billion dollars, approximately 6,800 million euros. The added value is in obtaining rigid and flexible packaging (recyclable, compostable, biodegradable, extend shelf life, antimicrobial and antioxidant).
Non-Food Packaging: Flexible and Rigid	Total volume for market packaging of USD 4.78 billion in 2019, reaching USD 22.38 billion in 2025, CAGR of 22.68 between period (2020 - 2025) Data June 2019. The added value is presented for manufacturers of rigid and flexible packaging. For now, 53% applications are rigid.
Cellulosic: Reinforce composites, textile, hygiene, industrial applications	The global cellulose fibre market size was valued at USD 20.61 billion in 2015. The market is expected to witness a significant growth on account of growing demand for environment-friendly, versatile, biodegradable, and cost-effective fibres globally. These fibres are produced from an ester of cellulose, which is mainly extracted from wood, bark or leaves of plants, and from other natural plant-based products. Industrial segment is expected to growth at a CAGR of 8.1% from 2016 to 2025. These fibres are widely used in the nonwoven, enzymes, plastic, paper, thermal insulation, and filtration industries.
Textiles: threads, PPE, Automotive, Agriculture,	The value in textiles applied in 2018 of 90 trillion dollars. The forecast is increases of 2-3% CAGR per year. The added value is to improve mechanical properties and sustainability and biodegradability as well as the consumer experience.
Non-woven: Wipes, Hygienic	The global global consumption of nonwovens: 11.2 million tons, 46.8 billion euros in 2019. With a growth rates for 2019–24: 6.7%. The added value for sutures to improve control and bacteriostatic degradability.
3D Printing: Prototyping, Functional part manufacturing, consumer goods	The global 3D printing market size is estimated to be USD 9.9 billion in 2018 and is expected to reach USD 34.8 billion by 2024. Factors such as ease in development of customized products, reduction in manufacturing cost and process downtime, government investments in 3D printing projects, and development of new industrial-grade 3D printing materials are driving the growth of the 3D printing industry. Currently, the trend in the 3D printing applications is shifting from prototyping to functional part manufacturing in various verticals, such as automotive, medical, aerospace, and consumer goods.

[39] https://www.european-bioplastics.org/market/
[40] http://news.bio-based.eu/2018-was-a-very-good-year-for-bio-based-polymers-several-additional-capacities-were-put-into-operation/
[41] https://www.marketsandmarkets.com/Market-Reports/metal-nanoparticle-market-138262033.html
[42] https://www.marketresearchstore.com/news/global-microencapsulation-market-165
[43] https://www.european-bioplastics.org/market/

Polymer Foams: Packaging, automotive	Global market of 113.9 trillion dollars in 2019, (105,000 million euros), a CAGR of 3.8% being projected from 2020 to 2027. The added value is in the development of new grades of polymers with improved performances: thermal, mechanical and better thermal and sound insulation.
Coatings: Paper rolls	The paints and coatings market is projected to grow from USD 153.94 Billion in 2019 to USD 199.88 Billion by 2024, at a compound annual rate of 5.4% during the forecast period[44].

2.1.2 Expected Barriers

Despite the opportunity of BIONANOPOLYS OITB and the capabilities of the consortium, there are external factors and barriers that may have an adverse effect on the long-term sustainability of the test bed and associated impact just described. Table 4 describes these barriers and how we will overcome it. Technical and management risks associated to the workplan are described in section 3.2.

Table 4: Expected barriers that may have an adverse effect on the sustainability of BIONANOPOLYS Test Bed.

Barriers	How BIONANOPOLYS will overcome it
Acceptance of new technology by the industry and products by customers	The access to the pilot lines with the experts will approach the technologies to users. Additionally, extensive training will be performed to offer consumer the best use experience
Access to secondary raw materials: Seasonal sustainability of feedstocks	Logistics in regards with storage and supply of biobased streams require specific attention. Certain organic byproducts are characterized by seasonal availability whereas also location, stability and product consistency can require specific requirements per product type. In order for the correct matchmaking to occur between byproducts & waste streams being produced on one hand and the need for such streams from the pilot partners, dedicated attention will be paid on storage and stable preservation. Depending on the type of product and processing pathway requirement, this can consist out of cold storage (freezing), dried storage or other types of preservations (e.g. fermented storage making use of ensiling and associated bacteria). UGENT coordinates the following initiatives and will make use of them for these purposes: BIOREFINE cluster, NUTRICYCLE VLAANDEREN (official regional nutrient recovery platform in the region of Flanders) and RE-SOURCE.BIO (International Network dealing on bioresource recovery – EU & transcontinental partnerships).
Price competition and market	A study of market process will be done, to cover other test beds and also universities, research centres, etc, in order to fix the best competitive prices
Difficulties to attract public and private investment	EBAN will be holding training methods with investors and Pitching Events. Special emphasis will be made in the communication campaign to ensure a significant number of attendances.
Very heterogenous range of products and clients	Within the project several partners are covering different technologies, and owing feedback in different materials and applications reaching a wide range of products and sectors to be applied. Clustering with other test bed will push to increase even more this fact.
Data sharing, collaborative business models & operational actions when developing the legal entity	Background IPR and unprotected innovation will be clearly identified under the leadership of G&S by the beginning of the project. Royalty-free licenses will be the base model to increase data sharing, while confidentiality will be covered by NDAs drafted with the support of G&S when necessary. A dedicated IPR and unprotected innovation committee will closely monitor these legal aspects, support other tasks, and anticipate potential legal issues by safeguarding at the same time the aims of open science and more specifically open data.
Be aligned with related standardization	BIONAOPOLYS will interact with standardization organizations such as CEN/TC 249/WG 9, ISO/TC 61/SC 14, CEN/TC 261 or ISO/TC 122 presenting the project results to be aligned with them. Communication tools and channels will be adapted to the needs of the standardization bodies in order to facilitate the exchange of information: good practices documentation, webinar, physical workshops, data exchange through the BIONANOPOLYS platform, etc.

2.2 Measures to maximise impact

[44] https://www.marketsandmarkets.com/Market-Reports/paint-coating-market-156661838.html

a) Plan for the dissemination and exploitation of the project results

The exploitation and dissemination of the project results is a key commitment for the consortium. A proper exploitation and dissemination of the project will ensure that it meets the multiple expected impacts mentioned in the topic. Furthermore, it is an ambition of the consortium members to transform the results into a marketable product, such that the partners are willing to invest time and resources in a project they believe has a strong impact and economic potential.

The policy and rules for protection, exploitation and dissemination activities are based on the principles set out in the Annotated Model Grant Agreement (GA) and will follow the Consortium Agreement (CA).

- Main issues tackled in the CA will be confidentiality, ownership of results, join ownership, background, access rights, dissemination and use of results, in addition to the internal governance of the consortium (see 3.2.).
- Main Issues tackled within BIONANOPOLYS will be managed by a new legal entity form after the project.
- Knowledge management and protection strategy has been agreed by all consortium partners and will be further detailed in the Consortium agreement

- Collaboration agreements with large industries and SMEs, on a regular basis, looking for the fidelity beyond a specific development, to guarantee constant collaboration and income in the short and medium term.
- BIONANOPOLYS will promote the participation of their users in public funding schemes at local, regional and European level to help companies in getting access to our services through external funding schemes.

BIONANOPOLYS EXPLOITATION STRUCTURE

LEGAL ENTITY: BIONANOPOLYS will examine in detail different structures for the formation of the legal entity as the main exploitation pathway for the OITB single entry point (SEP). Detailed information of SEP in **Task 2.2**

The purpose of the grouping is to facilitate or develop the economic activities of its members by a pooling of resources, activities or skills. This is intended to produce better results than the members acting alone and BIONANOPOLYS consortium considers that this legal entity fits perfectly with the management strategy of an OITB.

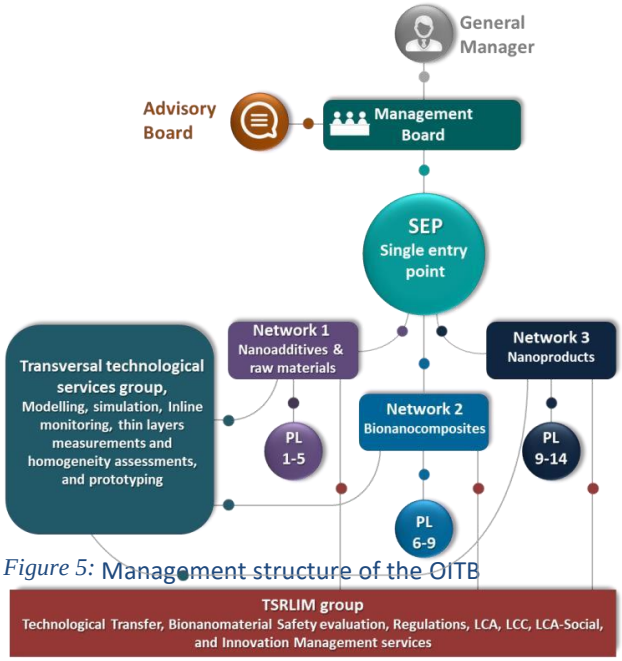

Figure 5: Management structure of the OITB

During M22 of the project a new legal entity will be formed to undertake the role of the BIONANOPOLYS SEP. The OITB members of the consortium have already agreed on the following management structure: **1.** A **Management Board** will be formed during the first stage of the project with the participation of 20 members, consisting of one representative from each OITB member (i.e. partners involved in pilot lines and in support services of the OITB. End users are not included). It should be mentioned that all entities will involve decision makers who have the capacity to represent their business. The board will be in charge of defining the OITB strategy and procedures, the workflows, the pricing conditions, business plans and IPR management, to overview the transversal technological services as well as to support OITB community building though workshops and dedicated events. The board will also make decisions related to the country that the SEP will be established, the legal form of the SEP, the selection procedure of the General Manager and any agreements among the OITB members, as well as between the OITB members and the customers. The management board will also supervise the process for the selection of the open call test cases, which will then be run under the supervision of the pilot line leaders. The SEP will be accessed by potential clients via an online platform where applicants will be able to fill out an application explaining in brief the innovative idea, they want to bring to the OITB. **2.** The **General Manager** will lead the SEP, being responsible to implement the strategy defined by the Board and manage financial issues of the SEP. The General Manager will chair the SEP and will be hired by the management board during the project.**3.** The **3 BIONANOPOLYS NETWORKS**, representing the groups of i. Nanoadditives and raw materials, ii. Bionanocomposites and iii. Nanoproducts. Each of these networks will be represented by one person from each pilot line they consist of. **4.** The **transversal technological services group** will be responsible for the Modelling, simulation, Inline monitoring, thin layers measurements and homogeneity assessments, and prototyping. **5.** The **TSRLIM group**, leading services related to technology transfer and support on technical activities, bionanomaterial safety evaluation, regulations, LCA, LCC, LCA-Social, and innovation management services (feasibility studies, business plans, market analysis, patent mapping, IPR, exploitation plans, business angels, access to finance, investors) and **6.** Finally, an **Advisory Board** (AB) of experts in the field of nano-enabled bio-based materials will be formed. The advisory board will be in communication with the management board offering their knowledge and consulting on the setting up of the pilot lines and the implementation of use cases. The structure and the interactions between the different groups and boards of the OITB are also shown in Figure 6. According to predefined workflows the interested customers will send an application through a dedicated web-platform. After they file their request the required pilot lines and groups will be contacted and the application will be evaluated, in terms of feasibility, estimation of cost and timeframe of delivery, with the supervision of the Management Board. Non-Disclosure

Agreement will be signed both by end users and the management board and Procedures towards the implementation of the project will be followed.

OPEN CALL. This operational structure will be validated during the project with external users by means of an open call for early adopters. The Open Call will promote the services offered by the test bed and to collect valuable feedback with the aim to ensure its sustainability after the project. A portfolio of services will be created in a form of "restaurant menu" to SMEs and LEs outside of the consortium. The "restaurant menu" will include all technical services related to BIONANOPOLYS technologies and supporting services (LCA, regulatory, standardisation, safety, business model development, IPR monitoring etc.). The services will be free for Open Call tenderers and the project partners will charge personnel and material costs from the project which will not surpass more than 25% of the project budget. In return, the Open Call tenderers will be requested to provide feedback on their experience using the test bed ("the restaurant experience feedback") that will be used to improve the test bed functioning, test bed single entry point visual appearance and easiness of access, application and communication process, quality of provided services and the service offers that will be included in the "restaurant menu". This feedback will be necessary to make sure that services and functioning of the test bed are ready for the paying customers after the end of the project ("the restaurant opening"). Once the project is finished, "the improved restaurant menu" with included the final prices for the offered services (already pre-defined in the annex in section 4-5).

BIONANOPOLYS has already promote the OITB and its Open call. 27 entities (mainly SMEs and clusters) have already shown interested in being informed if the project is funded (letters of support attached in section 4-5) **IPR STRATEGY**

Law firm G&S will be the partner in charge of the IPR and unprotected innovation strategy (Task 9.4 Legal aspects and IPR). Due to the nature of its activities, G&S is neutral and therefore well-placed to provide advice without any risk of conflict of interests. **IPR** are a **key component for the successful exploitation and implementation** of the project objectives and apply to some of the project results. However, part of the background and foreground is either unprotected though it would be protected at a later stage (e.g. with the filing of a patent application) or unprotectable (because it is outside the scope of any form of IPR protection). It is commonly one of the grey areas of most research and innovation projects which, if unaddressed, could lead to potential disputes. Therefore, the strategy and services also integrate the dimension of unprotected innovation. Specific provisions in the consortium agreement between partners will guarantee the strict **confidentiality** of unprotected innovation in order **not to preclude further IPR protection**.

During the entire duration of the project, G&S will implement the IPR and unprotected innovation strategy of the project and provide IPR management and protection services to the consortium members and test bed clients. The **IPR strategy fully takes into the open access** requirements of the H2020 programme and policy objectives of the EU Commission on Open Science. The background IPR necessary to carry out the different tasks foreseen in this project have already been determined but, as the situation could evolve between the submission of the proposal and the start of the project, a final listing of background IPR and unprotected innovation will be annexed to the consortium agreement to be signed between partners if this project is funded. Partners and clients of OITBs will retain their ownership rights on their background IPR and unprotected innovation but will make it available to other consortium members on a royalty-free non-exclusive licensing basis to carry out the tasks in this project.

If any background IPR and unprotected innovation of a partner is necessary for the exploitation of the foreground by other partners and clients, G&S will coordinate the negotiations between these parties. The foreground of the project will be subject to the ownership and access rights provided in the H2020 rules. If there is any joint innovation, its ownership would be divided between partners and – if relevant – clients of OITBs as per their share in the contribution to the joint innovation or, if it is impossible to determine the share, per capita.

The IPR and unprotected innovation strategy is problem-solving oriented. IPR disputes are unfortunately common during and especially after the end of a project. Therefore, a flexible, fast and efficient dispute resolution mechanism will be put in place from the project start in order to avoid issues escalate and block the innovation dynamics. The dispute resolution mechanism rules will be adopted by M6 under the leadership of G&S and after approval of the IPR and unprotected innovation management committee.

CANVAS MODEL

BIONANOPOLYS will make sure that the benefits of the demonstrated products developed within the OITB will be recognized by the community and as consequence the products shall be inserted in the market by following a business strategy dedicated to them.

Key partners
1. Other Test Beds
2. Investors to obtain follow-on finance
3. Regulatory – EC Directorate General: CLIMA; ENV; EASME; GROW; SANTE; JRC; RTD.
4. Local and Regional R&D agencies

Key Activities
- Branding
- Upgrading pilot lines
- Integration of the pilot lines under the OITB and exploitation strategy through the SEP
- Definition and creation of a Legal entity

Key resources
- Cutting edge pilot lines.
- Extensive network
- International reference RTOs

Unique value
- SEP in different official languages, user-friendly and open at fair conditions.
- Legal entity
- In-line monitoring
- Mathematical modelling
- Rapid prototyping
- Safe-by-design
- Safety assessment
- Business accelerator service

Customer relationships
- Workshops and specific training
- Business Accelerator service.
- From cradle to grave product development
- Customer service

Channels
- Outreach activities and open call tenders. - Research communities
- Clusters.

Customer
RESEARCH COMMUNITIES

INDUSTRY
- Packaging
-Textile
-Paper
-Cosmetics
-Nonwoven
-Agriculture
-Coating
-Foams -
....

Cost structures
-Pilot lines upgrading and integration
-Expansion of the test bed
-Organisational costs
-Maintenance of infrastructure

Revenue streams
-Technical development
-Validation of solution
-Risk assessment
-Sustainability assessment -
Business accelerator

JOINT EXPLOITATION STRATEGY

Key exploitable results have been identified, and the best exploitation strategy has been defined for each type of results – amongst the available exploitation routes: direct sale, provision of services, licensing. This exploitation strategy will be reviewed during the project

Pilot plants exploitation (link to WP3-5): Common exploitation strategy is foreseen, and will be governed following the management procedure of the legal entity and SEP. The exploitation route foreseen for this results will allow open access to external users (first estimation of prices for each pilot plant is detailed in section 4-5), renting a specific equipment, supporting technically the clients, and availability of different processing techniques to NP producers and final users, through **licence agreements.** Partners involved in protected developments will receive royalties from these sales, preliminary estimated in the range 6-10% of sales revenues. Geographical coverage of the licensing agreements will be carefully established to avoid any conflict of interest with the revenue source.

Partner	Key Exploitable results
ITENE	KER1- Synthesis of block copolymer obtaining from different types of biomass (PP3) KER2- Functional nanocapsules with controlled particle size obtained by Spray Dryer (PP5) KER3- Modification and functionalization of advanced cellulose-based material for different applications (PP6)
AITEX	KER4- Nanobubbles for finishing and dyeing garments (PP11) KER5- Nanofiber-based coatings by electrospraying process (PP14)
CIDAUT	KER6 -In-situ polymerization and melt compounding (PP7) KER7- Multifunctional films and semi manufactured products process by calendaring (PP9) KER8- Advance characterization of nanocomposites and modelling. 3D printing of thermoplastic bio nano composites by FDM technology (PP12)

CTP	KER9 -CNF-polymer film production process (PP10) KER10- High concentration CNF formulations (PP2) KER11- 3D-CNF spray process (PP14)
CENTITVC	KER12- Bio-based nanocapsules, nanocomposites and nanoproducts for the delivery of active agents (PP5) KER13-Pilot plant for production of bio-based nanocapsules, nanocomposites and nanoproducts (PP5) KER15- Functional coatings developed by roll to roll slot die coating (PP14)
STFI	KER16- Fibre nonwoven production (PP11) KER17- Meltblown nonwoven production (PP11)
BPF	KER 18- Extensive pretreatment with wide range of biomass (PP1)
ACIB	KER19- Different treatments for extraction biomaterials from biomass (PP1) KER20- Production of metallic nanoparticles from biomass annual plants (PP4)
PWR	KER21- Production of metallic nanoparticles from biomass (PP4)
Biotrend	KER22- Production lactic acid, succinic and PHA scaling and process identification (PP1)

Transversal Technologies (link to WP6): Modelling, monitoring a composition technology will be exploited by each responsible partner with the aim of improving the productivity and verification of pilot plants. Common exploitation strategy is based in **maintenance and training services**. In addition, specific exploitation strategies are envisaged for each partner.

	Key Exploitable results	Exploitation strategy
IMT	KER23: Models for predicting properties of novel bio nanocomposites KER24: Simulation results on use-cases KER25: Methodology for input-output properties database connected to a neural network ready-to-use	- Training services for intermediate and end users of products fabrication within and outside OITBs - Simulation and modelling services for SMEs RTOs - Software as a Service (SAAS) for predicting effective transport and barrier properties
CEA	KER26: Thin layer measurement KER27: Homogeneity evaluation (Monitoring and characterization services)	A new method and an associated prototype device will be the starting point for adaptation to other applications and interested parties. If higher industrial demands, the method will be transferred to an industrial partner for further valorization.
IRIS	KER28: Tailored inline monitoring systems (hardware+ software+ calibration models) for bio nanocomposites dispersion and composition leveraging photonic technologies	1- proof of concept (off line at IRIS when applicable or inline at one of the pilot lines supported by the client), 2- partial payment on order of the system and 3- final share of the payment upon successful delivery and demonstration.

Transversal Services (WP8-9) *A service portfolio will be exploited commonly though the SEP for OITB clients by means of 1) a* **business development team** capable of reaching **licensing agreements** with NP producers and final users at an international level. 2) To create an **international distribution network** (with direct and indirect sales) for the commercialization of *BIO*NANOPOLYS services All the services are defined in section 4-5 (pricing list) and will be part of NANOPOLYS portfolio.

- **ITENE**: REACH, Nano safety Risk assessment and/or Food Contact approval by EFSA or FDA, End- of -life (recyclability, and biodegradability) evaluation.
- **PARTICULA**: Eco-design approach, LCA, LCC, SLCA, algae biomass supply.
- **UGENT**: Biomass logistics optimization and supply chain
- **AXIA**: innovation management, Patent, Mapping: Business Strategy
- **EBAN**, EBN: IR Program and Pitching services
- **G&S**: IPR, regulations, legal advice, and compliance

Final products (WP7): End users will exploit their products independently of the OITB following an independent exploitation strategy – mainly based on **direct sale**.

Ambrosialab: Cellulose nanofibers and active nanocapsules for cosmetic and personal care formulations
Ambrosialab will introduce advance biobased additives and materials to improve their cosmetic formulations in order to obtain a sustainable formulation for cosmetic and functional food with the aim of replacing chemically produced formulations and be

able to explore new market niches.

Hilos Técnicos San Miquel S.L. Biodegradable synthetic fibres for agricultural applications

Hilos Técnicos San Miguel S.L. Will produce biobased and biodegradable synthetic fibres for application in agricultural Applications. With the biodegradable product will be used to short term applications where the textiles are left in the fields and biodegradation will help to reduce maintenance cost and time. Other product will include nanocapsules with active compounds (agrochemical) to release in a controlled manner, to enlarge the active effect, protecting better the crops.

TEXTISOL: Nonwoven hygiene products

The project will let us to develop and evaluate new biobased nonwoven wipes gaining a new and better-performing product. Replacing oil-based materials will enable growth in current and future business and speeding up growth in the existing segments (wet and dry wipes, table top, fem hyg.) by expanding the current production capacity. The project will open new markets offering products that currently are not available in the company portfolio, which will be exploited though direct sales.

NOVAMONT: Innovative biodegradable and compostable bioplastics incorporating NMs with superior barrier and mechanical properties.

The market of biodegradable plastics is expected to increase to over 503000 tons in 2022 due to the recent restriction laws. Results will be exploited with the support of new certification schemes for the materials that incorporates Nano compounds valorizing the increased renewable content, the biodegradability and compostability of the final materials, and the enhanced properties (e.g. mechanical and barrier). **112 new job positions** are expected to be created every kton of bioplastics produced

DANIPACK: **Active packaging for food applications**

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DANIPACK will be able to incorporate advanced active packaging developed from natural sources, what will give the opportunity to offer biodegradable packaging with reduce number of layers and improving the recyclability and an extended shelf-life (respect to biodegradable base material) provided by the incorporation of active nanocapsules.

DS SMITH: 1. **Cellulose nanofibers reinforcements for paper-based materials, 2. Hydrophobic nanofibers with Water and grease barrier and Paper with active properties.**

SMITH will introduce biobased materials offering sustainable solutions without renounce to features such as price, durability, hydrophobic, barrier capacity. Satisfying its client needs. SMITH will exploit by direct sales three innovative solutions, increasing the sustainability and the economic features of their products

CELLMAT: Biobased packaging in the form of thermoformed extruded foamed boards or thermoformed beads

Cellmat will act as pilot plant but in addition will act as end user of the packaging, which will exploit through licenses.

LOGOPLASTE: Sustainable, circular, performant rigid packaging solution for food (sauces) and personal care applications.

LOGOPLASTE will strengthen its array of services in terms of providing packaging using alternative, more sustainable, better performant raw materials. As the market gets increasingly competitive and more demanding of sustainable raw materials, the knowledge gathered from this project will act as a key differentiator as a packaging development partner and fulfill the demands of the client through direct sales**.**

MARKET ANALYSYS:

Market share: in order that BIONANOPOLYS will be sustainable as a platform, our assumptions are the following: Considering a Total Available Market of €85B and an estimation of an investment in R+D of the 10% to develop new bionanomaterials (Serviceable Available Market €8.5B). In a conservative approach, BIONANOPOLYS will be sustainable by capturing of the Serviceable Obtainable Market between 3.74% the first year and 5.64% the year 5 (with a cumulative revenue of €145M). Our target market are SME companies and their effort to invest in R&D is limited. That is the reason why our first approach is cautious. During the Project execution we will carry out a deep Feasibility Study about target markets and the potential of our project. Within this conservative approach we will be able create directly 507 jobs created in five years into the organisations involved in BIONANOPOLYS. Considering 1 job each €100K of net income.

Market Share and Sales					
	2022	**2023**	**2024**	**2025**	**2026**
Bio-based materials total market (M€)	32,629	41,471	52,710	66,994	85,150
Nano-enabled bio-based materials market (M€)	3,263	4,147	5,271	6,699	8,515
Investment in R&D (10% of the market value) (M€)	326	415	527	670	852
Market Share (%)	3.76%	4.84%	5.26%	5.63%	5.67%
Revenues Pilot Plant	8,021,300	12,025,400	14,254,800	15,872,400	17,324,300
Revenues Services	2,987,500	6,141,000	10,729,000	18,000,750	25,393,000
Revenues Transfer Know-How	1,272,000	1,902,950	2,750,850	3,836,780	5,604,560
Total Sales (€)	12,280,800	20,069,350	27,734,650	37,709,930	48,321,860

Operational costs - pricing list: in section 4-5 it is shown an estimation of the costs involved in the exploitation of the pilot lines, the complementary services and the know-how. Considering a conservative approach. It is expected to increase progressively the use of the pilot lines from 30% until 80% in 5 years (considering 1800 hours available per year) as well as the different services also included

Financial Feasibility Study: A first financial feasibility study has been done considering BIONANOPOLYS services as a whole and an integrated legal entity. It concluded that this business opportunity will enable the consortium to have a promising impact on this project: bringing a yearly revenue of €48M by the 5th year of sales, cumulative sales of €**146M**, EBITDA of **€17M** in the fifth year of sales, a Net Present Value (10% discount rate) of **€23,718,987** and an Internal Rate of Return of **55%**. To calculate the income of the pilot lines we have considered an increasing use between 30% and 80% of their capability. It is expected that the part of the benefits generated during the first years permit increases capacities and services of the project along the years. Therefore, BIONANOPOLYS will be able to capture more market than the initial potential after the first years. A complete **P&L Summary** can be seen below.

P&L Summary

	2022	2023	2024	2025	2026
Total Revenues (€)	12,280,800	20,069,350	27,734,650	37,709,930	48,321,860

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P&L Summary

2022		2023	2024	2025	2026
COGS (20% of sales) (€)	2,456,160	4,013,870	5,546,930	7,541,986	9,664,372
Gross Margin (€)	9,824,640	16,055,480	22,187,720	30,167,944	38,657,488
Operating Expenses (30% of sales) (€)	3,684,240	6,020,805	8,320,395	11,312,979	14,496,558
EBITDA (€)	6,140,400	10,034,675	13,867,325	18,854,965	24,160,930
EBITDA (% on sales)	*50.00%*	*50.00%*	*50.00%*	*50.00%*	*50.00%*
Amortizations	1,200,000	1,200,000	1,200,000	1,200,000	1,200,000
EBIT	4,940,400	8,834,675	12,667,325	17,654,965	22,960,930
EBIT (% on sales)	*40.23%*	*44.02%*	*45.67%*	*46.82%*	*47.52%*
Corporate Tax (25%)	1,235,100	2,208,669	3,166,831	4,413,741	5,740,233
Net Income	3,705,300	6,626,006	9,500,494	13,241,224	17,220,698
New jobs	38	67	96	133	173

DISSEMINATION PLAN

BIONANOPOLYS dissemination activities aim to communicate the project impact and results and engage future OITB users. To achieve the project results, traditional communication and dissemination activities are not enough. Deep engagement processes and participatory strategies with a social innovation approach are considered key for project success, and as such are part of the core activities of BIONANOPOLYS. Table 5 summarizes our strategic objectives customized to each target group and the tools for reaching them.

Table 5: Preliminary dissemination and communication plan

Target groups	Strategic objective and message	Dissemination & Communication tool
SMEs and Industry from Automotive, Textile, hygiene, food, & packaging	Technological transfer services for the development, testing & replication of new business models. Open access and technical support regarding the functionalisation of bionanomaterials offering tailor made solution, including in-line monitoring, modelling and characterization to optimise process, resulting materials quality, reduce scrap and ramp up time	Stakeholder Platform, European Workshops, key conferences and fairs, Webinars, leaflets & newsletters, product portfolio, Open Access publications, social media. OPEN CALL
Investors, business angels, industry	To engage stakeholders. Connect industry with potential investors and business angels in the developed solutions.	Stakeholder Platform, Conferences, newsletters, Workshops, social media
Academia, scientific community	To engage scientific community and share scientific and industrial last progress in nanomaterials functionalisation. To contribute on the ongoing and future interaction among science, industry, and law.	Stakeholder Platform, Scientific Open Access publications, workshops, conference, social media.
EU and national institutions & standardization	To generate interest on and awareness of the limits that law put to the fulfilment of bionanomaterials. Contribution to update standards according scientific and industrial needs.	Direct contacts through formal letters and meetings; workshops; publications

General public	To increase understanding and changing behaviour regarding management and use of bionanomaterials. Communicate the expected impact for society, economy, and environment.	Workshops, events, touch, taste, see experiences, visits to demo sites, social campaigns (Social networks, videos, etc.),

The next sections detail the dissemination tools to be used during the project.

Table 6: Events and congresses

SECTOR	Partner	Events to be attended	When	Where
General	CENTI, ITENE, CIDAUT, CTP	ICCCCF 2022: 16. International Conference on Cyclodextrin Chemistry and Cosmetic Formulations. ICMSST 2022: 16. International Conference on Materials, Surface Science and Technology. NANOSPAIN 2022. GEP (Spanish	2022	CH DE SP
SECTOR Where	Partner	Events to be attended	When	Where
		Conference of Polymers and Biopolymers). TAPPI Nano 2022: International Conference on Nanotechnology for Renewable Materials		
Composites	All	ICCM 2022: 16. International Conference on Composite Materials. JEC World - Leading International Composites Show; MATCOMP, Biofach	2022 2023	UK. FR
Packaging and food	ITENE, NOVAM	ICFPPM 2022: International Conference on Food Packaging and Packaging Materials. Food Processing & Packaging Technology Exhibition. INTERPACK; European Bioplastics conference	2022, 2023	UK, IT, DE
Textile	STFI, AITEX	Aachen-Dresden-Dekendorf International Textile Conference. ICTCLT 2022: International Conference on Textile Coating and Laminating Technology. Techtextil Fair; INDEX (nonwoven); FILTECH (filtration)	2022 2023	DE FE, IE, DE, CH
Foam	CELL	SPE Foams Conference; Polymer Foams (conference); FOAMEXPO (trade fair)	2023	EU
Bioprocesses and bioeconomy	ITE, NOVA ACIB, AXIA	European Congress on Biotechnology, Conference on Industrial Microbiology and Microbial Technology, International Symposium on Biocatalysis and Biotransformation (BioTrans); ECOMONDO trade fair 2022	2022 2023	EU
Biomass	UGENT	European Biomass Conference (EUBCE)	2022	EU
Safety	STFI, AIT, ITE	A+A - Fair for safe and healthy working, 8th International Conference NANOSAFE, 10th International Conference on Nanotoxicology	yearly	DE FR
Processing & modelling	CEN, ITENE, IMT, IRIS	COMPOUNDING WORLD EXPO EUROPE; K fair; International Conference of Surfaces, Coatings and Nanostructured Materials-NANOSMAT; COMSOL conferences. European Materials Research Society (E-MRS). ACHEMA (focus on process optimisation, industry 4.0	2022-2024	DE, EU
Coating	CEN, ITENE	European Coatings Show – The gathering of the coatings and paint industry	2022	DE
Innovation	AXIA, EBAN	9th International Conference on Innovation Management, Entrepreneurship and Sustainability. EBAN Annual Congress. European Angel Investment Summit	2021-2022	TBD

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Publications: In the case of publications, all peer-reviewed scientific publications (papers) related to BIONANOPOLYS results will be open-access, to be done through self-archiving (green open access). When a scientific publication or journal of high impact might be identified and considered of interest for the project, gold open access will be considered. In that case, the publication will be deposited in the repository mentioned in the data management plan, respecting the embargo period established by the scientific publication or journal publishers. However, due to the nature of the project (coming from previous research, high TRLs, innovation character of the OITB) BIONANOPOLYS consortium has already detected a higher interest on professional press related to the end users such as Open Journal of Composite Materials (ISSN Print: 2164-5612), International Journal of Cosmetic Science, Food Packaging and Shelf Life, Textile Research Journal (ISSN 0040-5175 print); Nonwovens industry (ISSN 0163-4429), among others.

Liaison with ongoing and future OITBs and relevant projects: BIONANOPOLYS will stablish synergies and collaboration with other projects (task 11.4). Special attention will be made with other OITB in other to promote alliances and commercial collaboration.

Trade groups, JTIs, PPP, platforms, and associations. BIONANOPOLYS consortium will make use of platforms and association they take part such as BBI, SUSCHEM, Biorefine and packaging clusters, A.SPIRE, SECPHO & EPIC (photonics), European Technology Platform for logistics (ALICE) and for textiles (ETP) and national for packaging (PACKNET), EDANA (non-woven association), EURATEX (Textile Confederation), COFRAC (French certification body). Standardization committee and similar bodies will be also contacted such as AFNOR, UNE, DIN NA106-0205-01 AK (Nonwovens and nonwoven machinery), Focus Group on Nutrient Recycling (EIP-AGRI), Technical WG on Struvite, Biochar and Ashes (STRUBIAS).

Training methods with investors. EBAN will lead training physical workshops focussed on Investment Readiness (IR) topics and cover critical issues for preparing companies to raise funds. Successful angel investors in business venture and innovation and qualified coaches will be invited. Regarding online activities, EBAN will make use of its free of charge tool **Start-up Includer**, an online deal room where entrepreneurs can securely upload information about their SME for individual investors and mentors to review. The deal-room also enables investors to engage in conversations with one another and with entrepreneurs for mentoring or pitching purposes, as well as to discuss syndication opportunities

Pitching Events to facilitate access to finance and investor communities. Pitching Events will take place during the Annual Congress and the European Angel Investment Summit hosted by EBAN every year. In these occasions, more than 300 business leaders, business angels, VCs, entrepreneurs, accelerators, and seed funds gather to discuss trends and opportunities Since 2018, EBAN events leverage on a mobile app for scheduling one-to-one and matchmaking meetings. As such, Pitching Events provide BIONANOPOLYS participants multiple benefits: firstly, they allow founders and companies to network and introduce their value proposition to a wide audience of interested parties in the perspective of long-term successful partnerships; secondly, they provide real and immediate feedback about the appetite of the investor community for a given business; thirdly, they represent a material opportunity to directly raise funds.

Data Management:

BIONANOPOLYS will develop a Data management plan (task 1.3 in WP1). BIONANOPOLYS will use and generate relevant data. The plan will explain how and whether this data will be used, exploited, or made accessible for verification and re-use. This plan will increase **BIONANOPOLYS** research efficiency, will save time and resources in the long run and will enhance data security and minimize the risk of data loss (see Table 7).

Table 7: Key information about BIONANOPOLYS research data management.

Type of data to be generated / collected	WP2: legal and confidential data to form the legal entity WP3: biomass and nanomaterials analysis data, pilot processes' specifications, nanoparticles physical and chemical properties. WP4: parameters process, in- line monitoring characterization, polymer characterization, and suspension characterization WP5: Product configuration, materials composition, conversion processes WP6: Analysis data (mathematical modelling, optical records, safety protocols) WP7-10: legal, market, administrative data from end users/external users of OITB, Technical specifications. WP8: safety, regulation, economic and environmental data from industry, migrations data. WP2,8,11: Stakeholders contact. Activity data from end users and open call candidates
Standards to be used	CEN/TC 17- Units of measurement; CEN/TC 42 - Certification of products; DIN EN ISO 3251 - content of involatile fractions; DIN EN ISO 2431 – flow time; CEN/TC 261 - Packaging, coatings; CEN/TC 411 - Bio–based products; EN 13432 - Biodegradability and compostability; EN 13430 and CEN/TR 13688 – Recyclability; CE/1935/2004 and EU/10/2011 (plastic) materials in food contact applications; CEN/TC 182 - safety and environmental requirements
Data exploitation. Accessibility for re-use and verification	Those data that are confidential (such as contact data for stakeholder or confidential information from all the partners linked to BIONANOPLYS) will be preserved with access restricted only by project partners. all data exchange transactions will be made in accordance with the EU General Data Protection Regulation (GDPR) Non-confidential data will be published as part of the different reports foreseen in the project and be deposited in the repository https://zenodo.org . https://dmponline.dcc.ac.uk will be used to create, review, and share data management plans
Data preservation	Making back-ups is essential. Data will be curated and preserved in a special repository allocated to the project, which will be stored in project coordinator servers and accessible only to project partners through the private section of project website.

b) Communication activities

The communication activities of BIONANOPOLYS will focus on ensuring a smooth information flow between the project partners and all relevant stakeholders and on raising awareness of the project activities and key results in a targeted and cost-efficient manner (more detail in WP11).

Core tools from the communication plan (Table 8) will be enhanced though the building up of the **BIONANOPOLYS Stakeholder Platform**. This is an interactive, tailored and user-friendly platform to compile all the information gathered and co-created related to the implementation of the innovative nanomaterials value chains (including new business models, results from the workshops, relevant news, partial and final open access results, media, regulatory aspects, action manuals (national/international replicability step-by-step manuals), and other relevant content generated as a consequence of BIONANOPOLYS). This Platform will be accessible to all internal (project members, advisory board) and external (advocacy and policy groups, citizens, sectorial associations, etc.)

Associated with document Ref. Ares(2020)6389918 - 05/11/2020
stakeholders. In essence, these will allow, on the one hand to increase transparency, provide data and perform and engagement, and on the other hand to maximize a wider successful dissemination of project results, it will be the first step to implement a SINGLE-ENTRY POINT. Due to the relevant and wire coverage of the communication plan, it will be led by ACIB and supported by one partner at national level. Those countries with more than one partner:
Spain-ITENE, France-CTP, Italy-Novamont-Italy, Belgium- EBN, PortugalAssociated with document Ref. Ares(2020)6389918 - 05/11/2020- CENTI, Germany - STFI. Communication will take place during the entire project and after it through the project stakeholder platform.

Table 8: Communication plan

Strategy	Target group	Means of communication	Coverage	KPI
Information on project, public deliverables	industry, policy makers, stakeholders	project website	International	30000 visits and 500 downloads per public deliverable one year after the project's end
Project's activities	Stakeholders and potential clients	6-monthly Newsletter, social media (facebook page, linked-in groups, twitter hashtag)	International	300 registered mails, 500 likes on facebook page, 50 linked-in posts and 50 twitter tweets per year
Achievements	General public, Investors	Two youtube promotional videos, leaflets	Europe	1000 views per video in 12 months from release
Lessons learnt	advisors	handbook	Europe	200 downloads in one year after the project's end
Publications	Researchers	High impact journals	International	Six publications and six citations in four years

3. Implementation

3.1 Work plan — Work packages, deliverables

BIONANOPOLYS Consortium has organised the work by creating three networks connecting the pilot plants dedicated to obtaining biobased nano -enabled products (Figure 2). These networks are supported and assessed by different activities, which aim to ensure their proper operation and the achievement of the project's objectives. Thus, to coordinate these activities, a set of Work Packages (WP) have been defined. A summary of these WP is shown in Figure 7 and the schedule in the GANTT chart below.

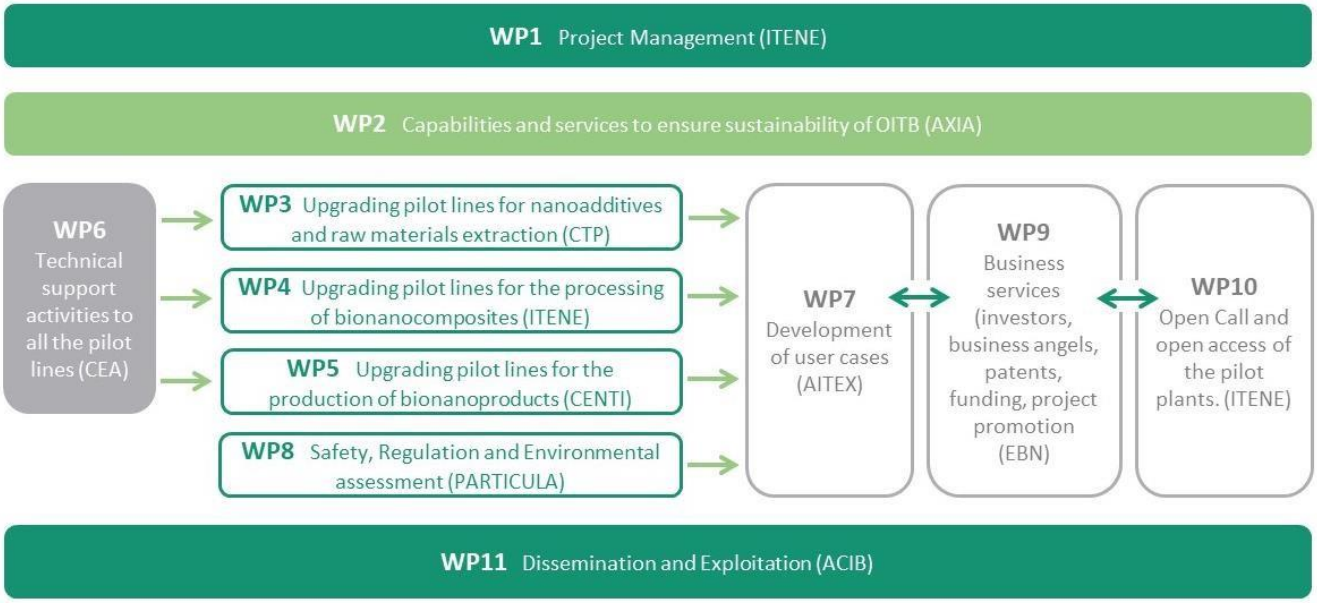
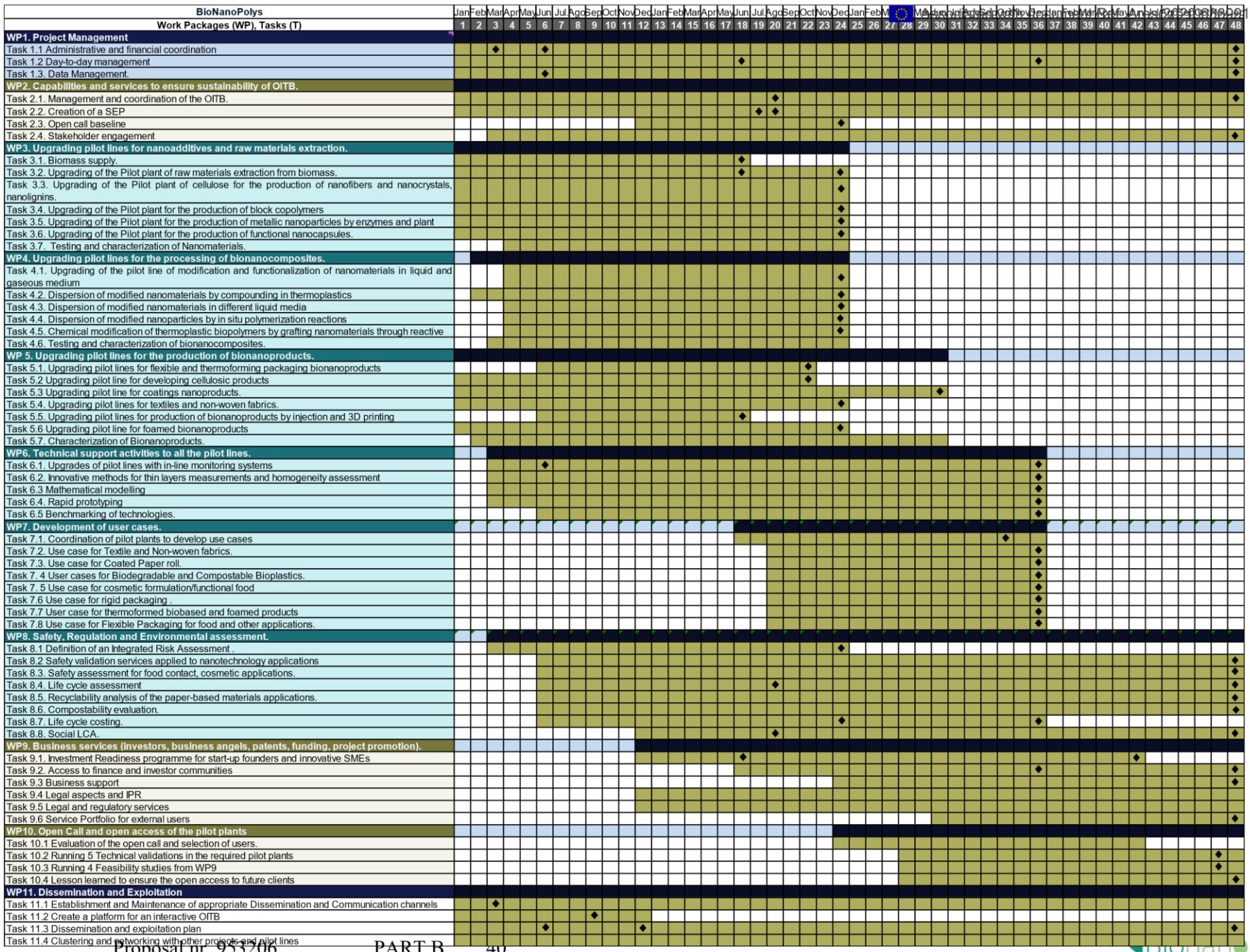


Figure 7. BIONANOPOLYS work package structure.

A list of the different Work Packages and a detailed description of each are shown below.

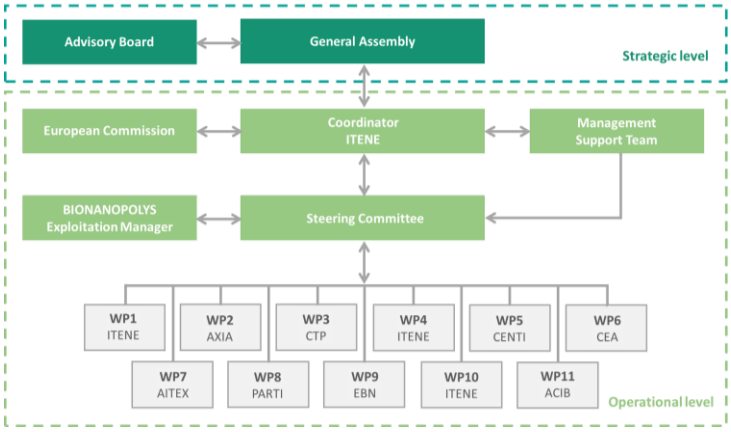
Table 9. Work package tittle table, including WP leader, stat and End month and Person Months associated.

WP	Work Package Title	Name	Person Months	Start Month	End month
1	Project Management.	1/ITENE		1	48
2	Capabilities and services to ensure sustainability of OITB.	19/AXIA		1	48
3	Upgrading pilot lines for nanoadditives and raw materials extraction.	4/CTP		1	24
4	Upgrading pilot lines for the processing of bionanocomposites.	1/ITENE		2	24
5	Upgrading pilot lines for the production of bionanoproducts.	5/CENTI		1	30
6	Technical support activities to all the pilot lines.	11/CEA		3	36
7	Development of user cases.	2/AITEX		18	36
8	Safety, Regulation and Environmental assessment.	15/PARTI		3	48
9	Business services (investors, business angels, patents, project promotion).	17/EBN		12	48
10	Open Call and open access of the pilot plants.	1/ITENE		24	48
11	Dissemination and Exploitation.	8/ACIB		1	48


BioNanoPolys
Work Packages (WP), Tasks (T)
WP1. Project Management
Task 1.1 Administrative and financial coordination
Task 1.2 Day-to-day management
Task 1.3. Data Management.
WP2. Capabilities and services to ensure sustainability of OITB.
Task 2.1. Management and coordination of the OITB.
Task 2.2. Creation of a SEP
Task 2.3. Open call baseline
Task 2.4. Stakeholder engagement
WP3. Upgrading pilot lines for nanoadditives and raw materials extraction.
Task 3.1. Biomass supply.
Task 3.2. Upgrading of the Pilot plant of raw materials extraction from biomass.
Task 3.3. Upgrading of the Pilot plant of cellulose for the production of nanofibers and nanocrystals, nanolignins.
Task 3.4. Upgrading of the Pilot plant for the production of block copolymers
Task 3.5. Upgrading of the Pilot plant for the production of metallic nanoparticles by enzymes and plant
Task 3.6. Upgrading of the Pilot plant for the production of functional nanocapsules.
Task 3.7. Testing and characterization of Nanomaterials.
WP4. Upgrading pilot lines for the processing of bionanocomposites.
Task 4.1. Upgrading of the pilot line of modification and functionalization of nanomaterials in liquid and gaseous medium
Task 4.2. Dispersion of modified nanomaterials by compounding in thermoplastics
Task 4.3. Dispersion of modified nanomaterials in different liquid media
Task 4.4. Dispersion of modified nanoparticles by in situ polymerization reactions
Task 4.5. Chemical modification of thermoplastic biopolymers by grafting nanomaterials through reactive
Task 4.6. Testing and characterization of bionanocomposites.
WP 5. Upgrading pilot lines for the production of bionanoproducts.
Task 5.1. Upgrading pilot lines for flexible and thermoforming packaging bionanoproducts
Task 5.2 Upgrading pilot line for developing cellulosic products
Task 5.3 Upgrading pilot line for coatings nanoproducts.
Task 5.4. Upgrading pilot lines for textiles and non-woven fabrics.
Task 5.5. Upgrading pilot lines for production of bionanoproducts by injection and 3D printing
Task 5.6 Upgrading pilot line for foamed bionanoproducts
Task 5.7. Characterization of Bionanoproducts.
WP6. Technical support activities to all the pilot lines.
Task 6.1. Upgrades of pilot lines with in-line monitoring systems
Task 6.2. Innovative methods for thin layers measurements and homogeneity assessment
Task 6.3 Mathematical modelling
Task 6.4. Rapid prototyping
Task 6.5 Benchmarking of technologies.
WP7. Development of user cases.
Task 7.1. Coordination of pilot plants to develop use cases
Task 7.2. Use case for Textile and Non-woven fabrics.
Task 7.3. Use case for Coated Paper roll.
Task 7. 4 User cases for Biodegradable and Compostable Bioplastics.
Task 7. 5 Use case for cosmetic formulation/functional food
Task 7.6 Use case for rigid packaging .
Task 7.7 User case for thermoformed biobased and foamed products
Task 7.8 Use case for Flexible Packaging for food and other applications.
WP8. Safety, Regulation and Environmental assessment.
Task 8.1 Definition of an Integrated Risk Assessment .
Task 8.2 Safety validation services applied to nanotechnology applications
Task 8.3. Safety assessment for food contact, cosmetic applications.
Task 8.4. Life cycle assessment
Task 8.5. Recyclability analysis of the paper-based materials applications.
Task 8.6. Compostability evaluation.
Task 8.7. Life cycle costing.
Task 8.8. Social LCA.
WP9. Business services (investors, business angels, patents, funding, project promotion).
Task 9.1. Investment Readiness programme for start-up founders and innovative SMEs
Task 9.2. Access to finance and investor communities
Task 9.3 Business support
Task 9.4 Legal aspects and IPR
Task 9.5 Legal and regulatory services
Task 9.6 Service Portfolio for external users
WP10. Open Call and open access of the pilot plants
Task 10.1 Evaluation of the open call and selection of users.
Task 10.2 Running 5 Technical validations in the required pilot plants
Task 10.3 Running 4 Feasibility studies from WP9
Task 10.4 Lesson learned to ensure the open access to future clients
WP11. Dissemination and Exploitation
Task 11.1 Establishment and Maintenance of appropriate Dissemination and Communication channels
Task 11.2 Create a platform for an interactive OITB
Task 11.3 Dissemination and exploitation plan
Task 11.4 Clustering and networking with other projects and pilot lines

3.2 Management structure, milestones and procedures

3.2.1 Project Management Structure



The management structure of BIONAOPOLYS is designed to achieve a clear definition of roles of each consortium member and to facilitate internal communication, monitoring of progress and an efficient use of project resources. This structure involves The Project Coordinator, The General Assembly, The Steering Committee, The Advisory Board, The Exploitation Manager and The Work Package Leaders and Task leaders.

PROJECT COORDINATOR (PC). The PC will chair the meetings of the Steering Committee. She will be responsible for managing the project in accordance with the description of action, consortium agreement and the project budget. She must identify any potential risks in the implementation of the project and putting in place immediate solutions in consultation with the relevant project management body. The PC must also ensure that the procedures put in place for overall quality assurance are optimal and complied with by project partners. PC will act also as scientific and technical leader and, together with the Steering Committee, directly coordinate the work of all Work Packages leaders. The PC must have a general overview of the entire project and be able to ensure compliance by all partners to project plans and agreements. The PC will be assisted in day-to-day management by a Management Support Team (formed by colleagues from the same institution). This Team will prepare and distribute the agenda and minutes of meetings of the various management bodies and will collate and transmit all reports required for delivery to the Commission.

Dr. Maria Jordá (ITENE) has been appointed to this position. She has extensive knowledge and expertise in EU funding, project management and execution in the areas of packaging, biobased materials, and nanomaterials. The Management support team will be led by **Ms. Rebeca Aguado** (ITENE) with years of experience in H2020 projects.

General Assembly (GA). The G**A** comprises one representative of each consortium member and represents the highest decision-making body of the Consortium. GA will decide and approve all matters of policy or any action that may have a substantive impact on the implementation of the project. The roles and procedures of the GA are:

- To meet once a year to receive and review project reports and approve major decisions made by the SC and PC. It may also meet in extraordinary sessions if needed.
- Meetings of the GA will be convened by the appointed Chairperson, who will take and distribute minutes of the meeting. Proceedings of the GA will also be included in progress reports to the Commission.
- Meetings will be chaired by the Chairperson appointed by consensus or a vote if required. The Chairperson will be assisted by a Vice, who will also be appointed by the GA.
- A majority of members attending is required to form a quorum, although three-quarters (21 out of 27) of the membership (including voting by correspondence) will be required for a decision to be binding.
- Key decisions of the GA will be endorsed and approved, as far as possible, by consensus and, in case of the lack thereof, by a vote. Each attending participating member will have one vote. In case of a tie of votes the Chairperson will cast a deciding vote, although each member has the right to submit an alternative position or in the case of a grievance to seek redress through procedures laid down in the Consortium Agreement (CA).

Steering Committee (SC). The SC will be the main functional body of the consortium and will oversee and control all aspects of the project implementation, including quality control of project outputs and dealing with critical risks at system or strategic level. It will be composed of a scientific/technical representative of each consortium member and chaired by the Coordinator. The Steering Committee will also include a Project Manager, who will assist the Coordinator in day-to-day project activities. The roles and procedures of the Steering Committee are the following:

- Reviewing the progress made by each of the work packages and agree on the action plans for the next period. It will also receive and review the reports of the Coordinator and the Exploitation Manager.
- The Committee will meet every three months. Meetings can take the form of video conferencing or other forms of electronic meeting although one physical meeting must be held at least every six months. Some of the physical meetings will be timed to coincide with General Assembly meetings in order to reduce travel time and costs.
- All decisions of the Steering Committee will be made as far as possible by consensus or, if necessary, by a simple majority vote. Voting by proxy or by correspondence will be accepted. The Coordinator will have the casting vote in the event of an equality of votes. A quorum of half of the Steering Committee members will be required for a decision to be binding.
- Minutes will be taken by the Coordinator's management team and distributed to all partners, including those who are not members of the Committee

Advisory Board (AB). The AB will serve as a conduit for linkages with the biobased industries. Members of the AB will comprise selected representatives representing the target sector of the project (packaging, textiles, non-woven, automotive, etc). The AB will also include identified experts that can advise on the scientific and technological aspects of the project to ensure that the project results are of the highest scientific quality. Members of the AB will be identified during the Kick-off meeting and will be formed by M6 of the project and updated if required with new member each year. It is foreseen that the AB will meet in parallel with GA meetings, although it will retain continuous dialogue through teleconferencing and exchange of correspondence. The AB will select its own Chairman and meetings will be convened and recorded by the Coordinator.

BIONANOPOLYS will arrange clustering activities, during and after the project, with related projects and OITB, whose main representatives will be invited to be part of the AB.

Exploitation Manager (**EM**). The **EM** will ensure that the project results are protected and shared equitably to participants in terms of access. The Exploitation Manager will be guided by a jointly agreed Plan for Dissemination and Exploitation, which will be drawn up by the Consortium. The EM will also be guided by the procedures established in both the Grant Agreement and Consortium Agreement for Background and Foreground as well as access rights to third parties and IPR strategy. The EM will help drive (and manage) innovation within the project via a "commercial pull", in liaison with the members of the Steering Committee and the Advisory Board. The EM will also be responsible for capturing feedback to the consortium from the users of the OITB to improve the user interface and user experience.

Mr Anthony (G&S) has been appointed to this position. He works in the law firm G&S, being neutral and therefore well-placed to provide advice without any risk of conflict of interests. He has years of experience in legal aspects, compliance and IPR and unprotected innovation strategy.

Work Package Leaders (WP Leaders). The WP leaders are experts in the relevant subject areas of the project and will form the basic fundament upon which the implementation of project tasks will revolve. WP Leaders will ensure that the project implementation is based on agreed work plans contained and that *Milestones* are achieved in accordance with the **Description of Action (DOA).** Any deviations from or major changes to the work plans in the DOA will require prior approval by the Steering Committee and, in some cases, approval of the EC Project Officer. WP Leaders will collect and collate contributions from various Task Managers and prepare reports to be presented at Steering Committee meetings. They will also undertake detailed risk analysis of tasks in their WP and alert the Coordinator to any potential risks and contingency plans for averting them.

Within work packages, each partner institution will appoint a **Task Manager** for each task they are involved in. Task Managers will have direct responsibility for *Deliverables* under their Task, report to their respective WP Leaders and follow plans proposed by them. They will also have to report and discuss any technical problems with the WP Leader to find appropriate solutions

11 WP leaders compose this structure. Each WP leader have been chosen due their previous experience and sector.

MANAGEMENT STRUCTURE AFTER DE PROJECT:

BIONAOPOLYS Management structure will be aligned with the management structure of the legal entity to be stablish within the project duration. The management structure of the entity will be executed within WP2 and led by AXIA. The management structure and mechanism for its exploitation have been define in impact section in order to ensure the sustainability of the entity after the project.

3.2. Critical risks for implementation and risk management

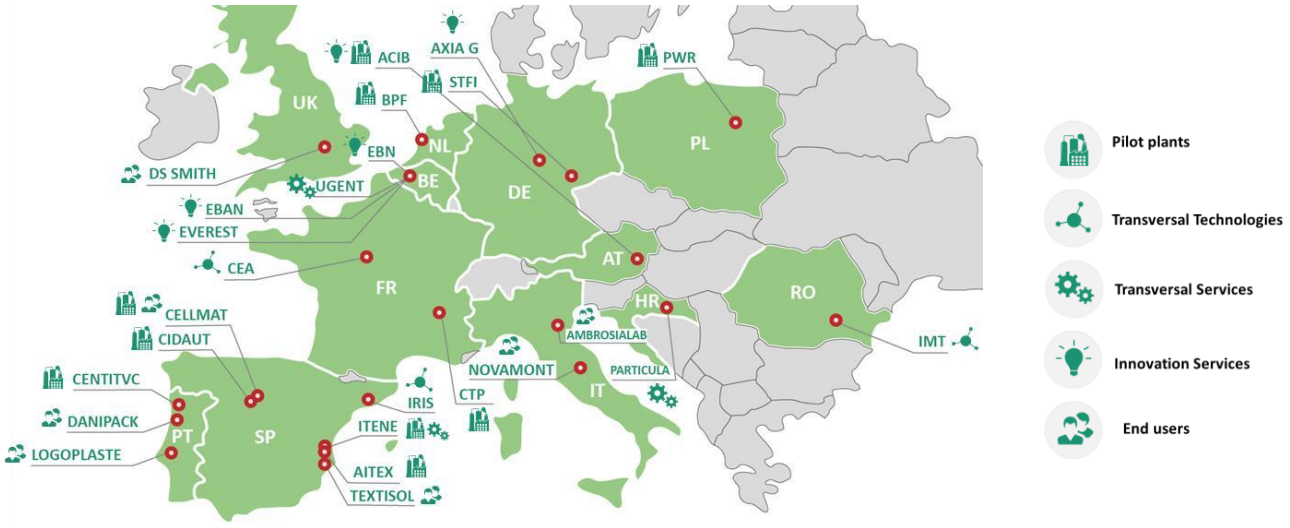
As part of the project management, risks assessment is an essential task; therefore, it will be part of the internal reports and will be discussed in all meetings to analyse the potential risks and to manage them. For these tasks BIONANOPOLYS counts with the following strategies to mitigate the possible risks:

Description of risk (indicate level of likelihood: Low/Medium/High)	WP	Proposed risk mitigation measures
Poor WP operation and communication leading to delayed/failed objectives (Probability: Low / Impact: Low)	WP1-WP2	Regular meetings will be held to enhance the sound communications tools and development of a tracking process among partners. Also, the preparation of implementation plans for any WP will be carried out. In case of failure, the project management procedures will be re-assessed
Partners with unique know-how or role within the project leave or underperform. (Probability: Low / Impact: Medium-High)	WP1-WP2	Seek early Steering Committee decision to re- allocate internal/external partners or invite external entities through official searching tools. The Consortium Agreement will describe the terms and conditions to accept a new partner in the consortium and its role in the OITB legal entity.
The supplied waste (raw material) is not sufficient for the development of the pilot trials due to seasonal availability.	WP3	Among all selected raw materials, other agricultural or food waste supplies will be considered in priority (especially for the food processing wastes available throughout the year). Smaller loading of wastes will be considered.
Low suitability of the synthesized nanomaterials for the dispersion in the matrices (Probability: Medium/Impact: Medium)	WP3	Nanomaterials will be characterized to assess if they meet the stablished requirements for each WP and for each preconized product. If necessary, the addition of commercially available bio-based nanomaterials will be taken in consideration as means of ensuring the project's goals.
Incapability of pilot lines upgrade/incorporation of inline monitoring systems (Probability: Medium/Impact: High)	WP3 WP6	The control and monitoring necessities of pilot lines will be assessed and its capacity to insert these new system components. The upgrade will consider all the process parameters involved in the development of the nanomaterial and the design of the equipment, which in turn will help define the design of the upgraded pilot plant and of the monitoring systems.
Poor dispersion of NMs leading to bionanocomposites with low performance. Prob. Low, Impact: High	WP4	Redesign of the bionanocompounds and modification of the conditions set in the upgraded pilot-plants. Reactive extrusion can also be used to improve NM.
Industrial scale-up of coating and compounding processes is not feasible from a environmental and cost point of view. (Probability: Low/Impact: High)	WP4 WP5	Technical feasibility of the developments and processes has been previously evaluated and demonstrated by the corresponding partners. WP4 and WP5 will specifically address scaling-up considering the previous results and will work on environmental, energetic and cost efficiency.
Final properties of bionanoproducts developed do not meet the set requirements (Probability: Medium / Impact: High)	WP5	Further evaluation and modification of nanoadditives developed in WP3 and adjustment of formulations on WP4 will be performed to enhance their properties and processability. An exhaustive optimization of processing parameters and synergies between partners working in the same PP will be established to achieve the expected properties.
Performance of standard XRF methods not meeting the expectations for coating measurements (Probability: Medium / Impact: Low)	WP6	Hardware improvements: search and test other detectors and/or sources, optimize energy range with filters, use a controlled atmosphere for lowenergy measures (e.g. inert gases). Method improvements: i) use optimized substrates; ii) use artificial intelligence methods for advanced spectral processing approaches.

Lack of correlation between simulation and service behaviour data of the printed nanocomposites (Probability: Medium / Impact: Low)	WP6	Redesign the testing methodology and improve test benches to reproduce main conditions. Collect new data sets with improved control of parameters and reassess the correlations.
Low printability of polymer nanocomposites due to water-absorbing trend and/or high crystallinity (Probability: Medium / Impact: Low)	WP6	Working on the equipment: re-design and propose modifications of the printer machine in order to allow controlling moisture and temperature along the entire printing volume. Working on the bio-based nanocomposites: propose and apply adapted pre-treatments (conditioning), optimized process and printing parameters and post-treatments (annealing).
Impossibility to scale-up the developments and technologies from the pilot scale to the industrial scale (Probability: Medium/Impact: High)	WP7	Synergies between the RTOs and technology providers will ensure the industrialization of our solutions and test beds. If necessary, contacts with other industrial players for the large-scale production of the nanomaterials and nanocomposites to ensure the end-users' production needs.
The products from the end-user cases do not meet the delineated	WP7	Development of a product that allows combining solutions developed in the WP3, WP4 and WP5, to achieve the best synergy between the

Description of risk (indicate level of likelihood: Low/Medium/High)	WP	Associated with document Ref. Ares(2020)6389918 - 05/11/2020Proposed ris mitigation measure
specifications and requirements (Probability: Medium/Impact: Medium)		nanomaterials, composite polymeric/resin matrix and the substrate. The industrial process will be studied, and the parameters will be adjusted to optimize the process and enhance the properties of the final products.
Lack of investment ready companies to take part in Pitching Events (Medium)	WP10	companies still in their inception stage will receive valuable recommendations to move on the next steps of their venture rather than be exposed to a jury expecting investment ready opportunities.
Lack of interest in the workshops and open-call (M)	WP9-10-11	Redefine very well the target sectors to approach and the message

3.3 Consortium as a whole



BIONANOPOLYS is formed by a consortium of 27 partners located across all Europe and representing 12 countries, 10 RTOs, 2 Universities, 10 SMEs, 3 Large industries and 2 business non-profit associations. ITENE has been chosen as a coordinator due to its experience as participant and coordinator of more than 80 European projects. Each single participant brings specific infrastructure, technologies, and skills (including humanities) required for an optimal execution of the project.

- 11 technological institutes/companies covering all areas require for the deployment and exploitation of the 14 Pilot plants. First Ugent will ensure the biomass to be treated by BPF (processing), ACIB (enzymatic treatments), BIOTREND (bioprocesses). Rest of partners represent key sectors: ITENE (packaging), AITEX (textile), CIDAUT (automotive), CTP (cellulose and lignin), CENTI (encapsulation)), STFI (non-woven), PWR (nanoparticles).
- 3 additional partners will enhance the capacity of the pilot plants through composing monitoring technologies (IRIS, CEA) and mathematical modelling (IMT).
- Environmental and safety performance will be ensured and validated by the PARTICULA and once again by ITENE, respectively, which also will make use of its compostability/recyclability accredited laboratory.
- 4 Additional companies will maximize the innovation capacity of the project: AXIA (management innovation), EBAN (investors), EBN (business) and G&S (law firm covering legal aspects and compliance). ACIB will work also on dissemination and exploitation due to its experience as exploitation and Communication manager.
- 8 industrial partners of different sizes, sectors and operational capacity (6 SMES and 2 large industry): Ambrosiolab (cosmetics), Textisol (textiles and non-woven), DS SMITH (cellulosic packaging), DANIPACK (metallized packaging), Logoplast (plastic bottles), NOVAMONT (polymers in several applications), HSM (technical threads). CELLMAT besides offering a pilot plant will act as validator of nanoparticles.

3.4 Resources to be committed

BIONANOPOLYS project is distributed in eleven work packages with a **total budget of 13.282.958,75€**, meanwhile asked **funding is 11.756.623,63€**. 8.353.788,00€ corresponds to personal cost and 2.220.659,00€ to equipment and other cost for upgrading the pilot plants and validate the products. Regarding gender issues, Chili is coordinated by a woman (Dr. María Jordá, ITENE). Moreover, in total 7 WPs are led by women, having the majority in vote decision. **The distribution of PMs** among WPs is proportionated an equilibrated. There are not big differences except for WP7, which require bigger efforts since the end users will be validating 14 pilot plants to develop all the products. This requires the involvement of them, but also from the pilot plants owners. End users will not be participating in the creating of the legal entity (WP2), pilot plants upgrading (Wp3-6) and open call. On contrary, pilot plants owner's activity within the development of the feasibilities studies (WP9) remains residual.

4.2.